GE VERNOVA

# Hazard Management

1.0.0

Installation Guide

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# Change History

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# 1. Introduction

This chapter provides an overview of the installation process and describes the conditions that must be met prior to installing Hazard Management System.

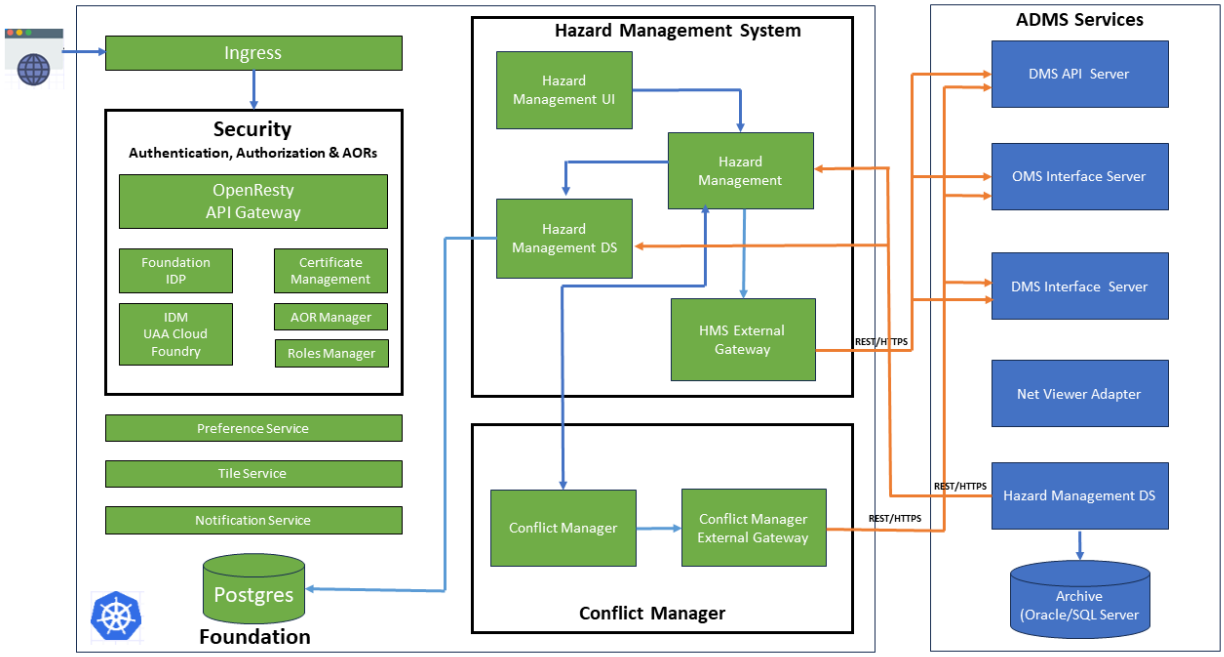
## 1.1. Hazard Management Software Overview

Hazard Management is a Kubernetes-based microservice architecture built on GE Vernova Foundation. Therefore, you must first create a Kubernetes environment before deploying Hazard Management. The application deployment is certified for the Kubernetes RKE2 cluster deployed using GE Vernova's Kubernetes K8s PDI. For details about installing the Kubernetes RKE2 cluster and deploying the Foundation solution, consult your GE Vernova project team.

Hazard Management uses an umbrella Helm chart for deployment. Helm helps define, install, and upgrade Kubernetes applications and manage Helm charts.

A simplified deployment includes the following steps:

- Create a Kubernetes environment using GE Vernova's Kubernetes PDI.
- Deploy the Foundation Base, Foundation OSB, and Foundation UI Server using Helm charts or ArgoCD.
- Install and configure ADMS.
- Deploy the Enterprise Switching umbrella Helm charts.



# 1.2. Preparing for Installation

## 1.2.1. Hardware Requirements

The HMS hardware requirements are driven by the Foundation hardware requirements.

To successfully install and use HMS with the demo system (Fantasy Island), the minimum Kubernetes Foundation hardware requirements are as follows:

- **Kubernetes Servers:** For performance reasons, use at least four nodes with the following properties:
  - **Memory:** 16 GB per node
  - **CPU:** 4 cores per node
  - **Storage:** At least 200 GB per node
  - **Operating System:** Foundation Base is supported on Red Hat Enterprise Linux (RHEL) 8.3 64-bit, using the Minimal Install option, with the latest updates and access to RHEL RPM repositories.

    > **Note**
    >
    > A high availability (HA) topology-capable cluster setup is highly recommended. The hardware configuration will vary depending on the container replica set required in the HA topology. To support a basic HA topology, use at least five nodes with the above hardware configuration, ensuring that:
    >
    > - Three of the nodes are configured as "stacked" master nodes (master plus agent)
    > - The remaining nodes configured as "agent" nodes
    >
    > For greater workloads and to support more concurrent clients, the resources assigned to each Storm Assist worker node can be increased.

- **Deployment Server:** A server specifically for deployment purposes.
  - **Memory:** 16 GB
  - **CPU:** 4 cores
  - **Storage:** 200 GB
  - **Operating System:** Red Hat Enterprise Linux (RHEL) 8.7 64 bit
- **HMS Clients:**
  - **Memory:** 4 GB
  - **CPU:** 2 cores
  - **Display:** Using a monitor with full HD (1920x1080) display resolution is recommended

The above sizing numbers reflect minimal resource requirements for Foundation layers; the actual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sizing should be determined for a specific solution and set of workloads, with realistic data sets (and load test).

It is recommended that you consult with GE regarding your project-specific hardware and software requirements.

## 1.3. Operating System

This section provides compatibility details for operating systems.

- **Kubernetes Server Operating System:** Red Hat Enterprise Linux (RHEL) 8.7 64 bit
- **Deployment Server Operating System:** Red Hat Enterprise Linux (RHEL) 8.7 64 bit
- **Client Operating System:** HMS clients are platform-independent.


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# 2. System Overview

Hazard Management System (HMS) is a Kubernetes-based microservice architecture built on .NET 8.0. The system manages hazardous events, safety zones, and electrical grid operations through a set of containerized microservices. Therefore, you must first create a Kubernetes environment before deploying the Hazard Management System. The application deployment is certified for Kubernetes RKE2 clusters deployed using GE Vernova's Kubernetes K8s PDI. For details about installing the Kubernetes RKE2 cluster and deploying the Foundation solution, consult your GE Vernova project team.

Hazard Management System uses an umbrella Helm chart for deployment. Helm helps define, install, and upgrade Kubernetes applications and manage Helm charts.

A simplified deployment includes the following steps:

- Create a Kubernetes environment using GE Vernova's Kubernetes PDI.
- Deploy the Foundation Base, Foundation OSB, and Foundation UI Server using Helm charts or ArgoCD.
- Install and configure ADMS.
- Deploy the Hazard Management System umbrella Helm charts.

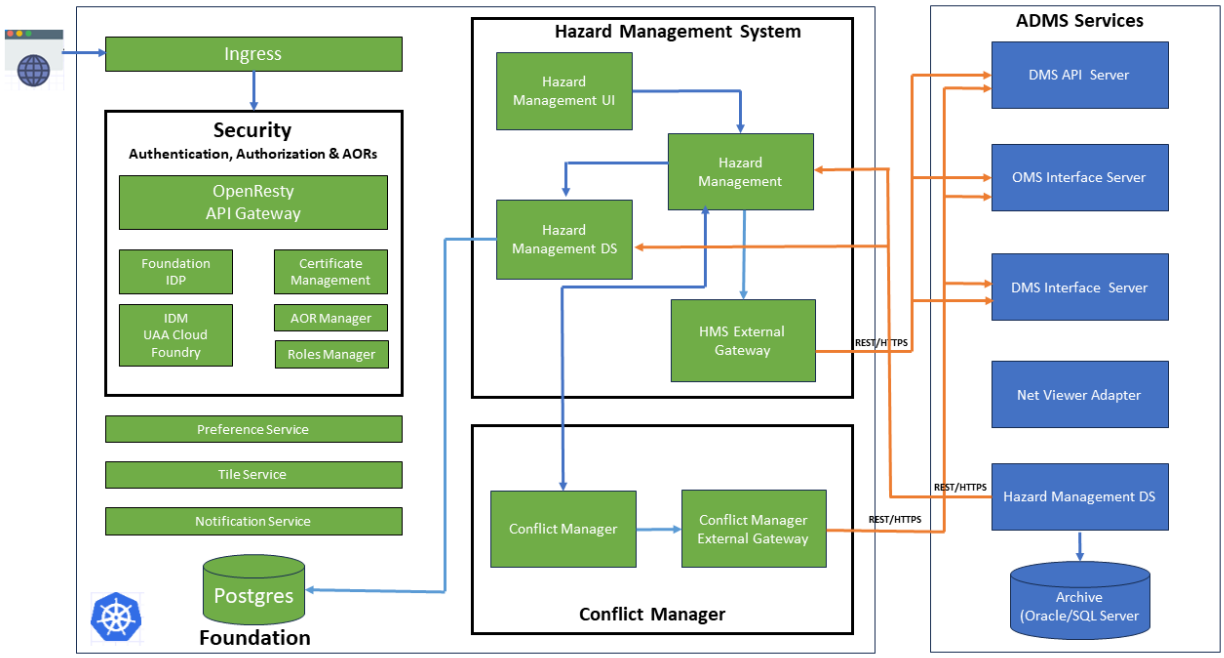


*Figure 1. Hazard Management Key Components*

## 2.1. Foundation Components

The Foundation platform provides essential services to support applications. HMS leverages the

following key Foundation services:

- **Foundation Base Services:** Includes common foundation services:
  - **Security:** Offers features like user authentication and authorization, including the UAA server and AOR Manager.
  - **Elasticsearch:** Stores application logs in a persistent Elasticsearch database.
  - **External Gateway:** Facilitates secure communication between HMS and ADMS systems.

  The Foundation Base Services layer also includes alerting, monitoring, and routing capabilities.
- **Foundation OSB Services:** Provides integration services:
  - **Notification Service:** Uses Kafka to send event state updates to the Hazard Management UI via WebSocket.
- **Foundation UI Server Services:** Includes UI services required for HMS:
  - **Preferences Service:** Saves user preferences and configurations in the Postgres database.
  - **Tile Service:** Retrieves tiles and configurations from the ADMS NetViewer Adapter and provides the data to the HMS Network Viewer.

For more details on Foundation services, refer to the following:

- **Foundation Software Installation:** Covers installation, configuration, and maintenance of Foundation services.

## 2.2. Hazard Management System

HMS includes the following services and custom resources:

- **hazard-management-ui:** Service containing the HMS user interface.
- **hazard-management:** Service containing the business logic to process Hazard Events.
- **hazard-management-ds:** Service to manage read/write to database. This can be configured to connect with Postgres, Oracle and SQL Server. This service is also deployed on the ADMS machine to write to the ADMS Archive database.
- **hms-external-gateway:** Supports failover capabilities and secure communication to ADMS servers.
- **External Gateway Routes**
  - **nva-main-eg-route:** External Gateway route for connection to the NetViewer Adapter
  - **ois-eg-route:** External Gateway route for connection to the OMS Interface Server.
  - **dms-api-eg-route:** External Gateway route for connection to the DMS API Server
  - **dis-eg-route:** External Gateway route for connection to the DMS Interface Server.

The HMS UI supports a thin client, where the client is simply a web browser that connects to the


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hazard management UI service. The HMS UI is implemented using Angular. The backend services, Hazard Management, Hazard Management DS, are implemented in .NET Core.

## 2.3. Conflict Manager

Conflict Manager includes the following services and custom resources:

- **conflict-manager-service:** Service that identifies conflicts between Hazard Events, Switching Orders, and Planned Outages within a specified time period.
- **conflict-manager-external-gateway:** Supports failover capabilities and secure communication to ADMS servers.
- **External Gateway Routes**
  - **ois-eg-route:** External Gateway route for connection to the OMS Interface Server.
  - **dms-api-eg-route:** External Gateway route for connection to the DMS API Server
  - **dis-eg-route:** External Gateway route for connection to the DMS Interface Server.

The Conflict Manager is built using .NET Core and has no user interface. Other applications, such as Hazard Management and Enterprise Switching, interact with Conflict Manager services to identify conflicts.

Currently, Conflict Manager is deployed alongside the Hazard Management services. In future releases, it will be deployed in a separate zone for improved scalability and isolation.

## 2.4. Hazard Management System Integration with ADMS

Hazard Management uses the following ADMS Services

- **DMS API Server:** A web server interface that enables third-party applications to communicate with the ADMS DNAF server (DnafSrv) through REST APIs. Hazard Management uses the DMS API Server to load network data and apply restrictions.
- **OMS Interface Server:** A web server interface that facilitates communication between third-party applications and the ADMS OMS server (NetServer) using REST APIs. Hazard Management utilizes the OMS Interface Server APIs to create or update planned outages in the ADMS for specific hazard events.
- **DMS Interface Server:** A web server interface that allows third-party applications to interact with the DMS server (NetSrv) via REST APIs. Hazard Management relies on the DMS Interface Server APIs to place and remove tags.
- **NetViewer Adapter (Map Modes):** Provides network data to the Hazard Management System (HMS) network viewer.


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## 2.5. Conflict Manager Integration with ADMS

Conflict Manager uses the following ADMS Services

- **DMS API Server:** A web server interface that enables third-party applications to communicate with the ADMS DNAF server (DnafSrv) through REST APIs. Conflict Manager uses the DMS API Server to load network data.
- **OMS Interface Server:** A web server interface that facilitates communication between third-party applications and the ADMS OMS server (NetServer) using REST APIs. Conflict Manager uses the OMS Interface Server APIs to retrieve planned outages data from the ADMS for a specified time period.
- **DMS Interface Server:** A web server interface that allows third-party applications to interact with the DMS server (NetSrv) via REST APIs. Conflict Manager uses the DMS Interface Server APIs to retrieve Switching Order data from the ADMS for a specified time period.

## 2.6. Connection Security

A secure connection between the HMS and ADMS systems is established using an HTTPS connection with TLS termination.

For HTTPS with TLS, SSL certificates must be obtained from your certificate authority. The certification authority is based on Active Directory or other services. Therefore, you need to have two certificates: one for the Kubernetes Ingress Controller and another for ADMS interface servers.


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# 3. Installation Overview

For a list of certified and supported operating systems, as well as the required third-party applications, refer to the "System Requirements" chapter in the *HMS Release Notes*.

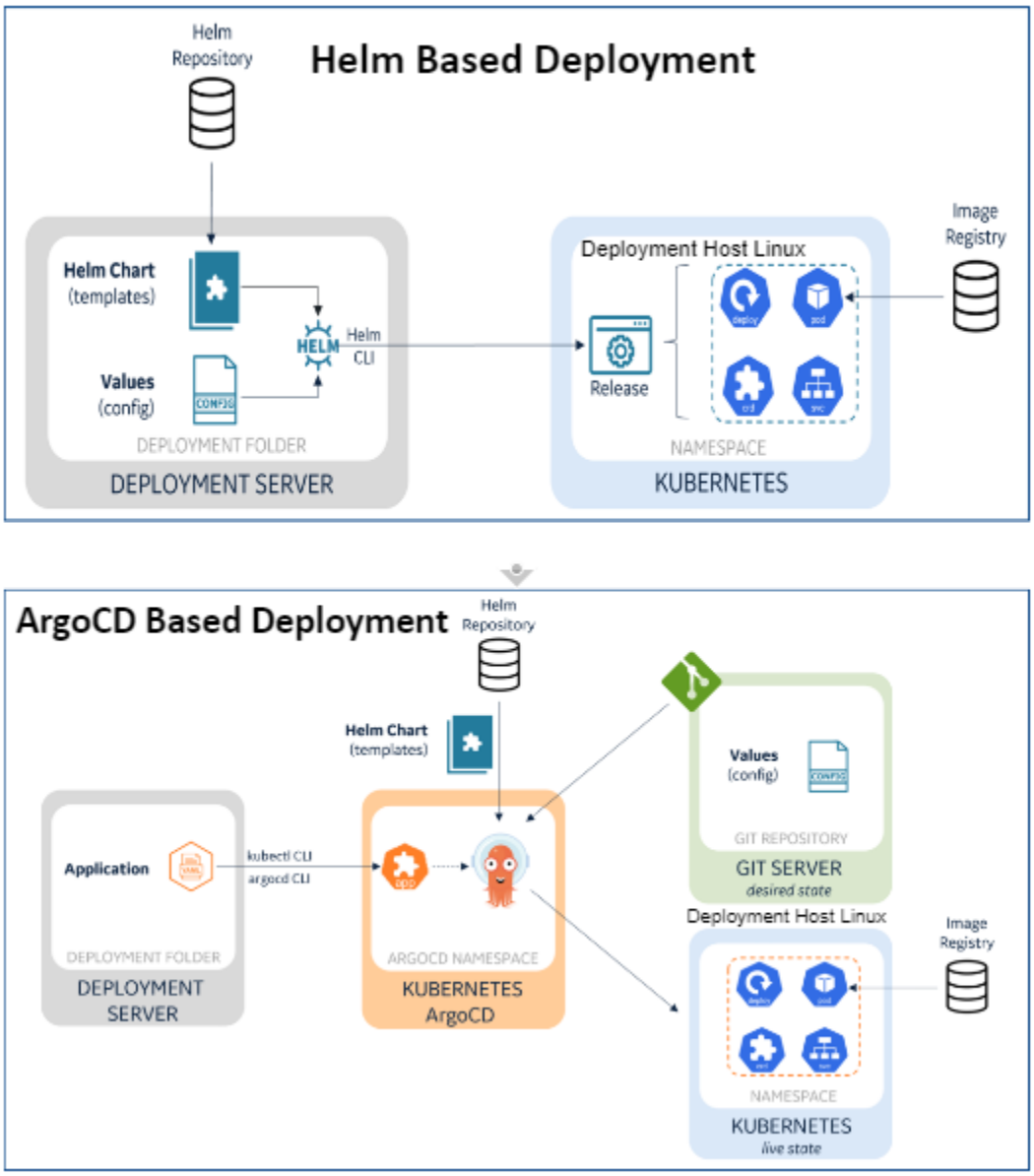


*Figure 2. Deployment Overview*

HMS components (containers) are orchestrated by Kubernetes. Therefore, a Kubernetes environment must be created before deploying HMS. The application deployment is certified for the Kubernetes RKE2 cluster, which is deployed using GE's Kubernetes PDI. For details about installing the Kubernetes RKE2 cluster and deploying the Foundation solution, consult your GE project team.

HMS uses GE's Deployment Model 2.0 for deployment, based on Helm or ArgoCD with Helm to the

Kubernetes cluster.

HMS utilizes Helm for templating deployments, helping define, install, and upgrade Kubernetes applications and manage Helm charts.

A simplified deployment includes the following steps:

1. Creating a Kubernetes environment using GE's Kubernetes PDI.
2. Deploying the Foundation Base, Foundation OSB, and Foundation UI Server Helm charts.
3. Performing pre-deployment steps for HMS.
4. Deploying the HMS Helm Chart.
5. Performing post-deployment steps for HMS.
6. Configuring ADMS.
7. Validating the HMS setup.

Detailed deployment information is provided in this document.

> **Note**
>
> Prerequisites for ADMS are outside the scope of this document. For more information, refer to the *ADMS Release Notes*.

## 3.1. Hardware and Operating System Prerequisites

For a list of supported hardware and operating systems, refer to the "Hardware Requirements" and "Operating System" sections in the *HMS Release Notes*.

## 3.2. Software Prerequisites

### 3.2.1. Installation Artifacts

Before installation, obtain the following artifacts from your GE project team:

- Cache containers with the required Docker containers (`hms-cache-16.18.0-18.tar.gz`)
- Helm charts for HMS (`hms-deployment-16.18.0.tgz`, `strimzi-kafka-cluster-16.18.0.tgz`, and `conflict-manager-deployment-16.18.0.tgz`)
- Sample input files for deployment (`k8s-base-manifest.yaml`, `foundation-base.yaml`, `foundation-osb.yaml`, `foundation-ui-server.yaml`, and `hms.yaml`)

### 3.2.2. Server

HMS server has the following software requirements:

- RKE2 Kubernetes (installed using GE's Kubernetes PDI)
- Foundation Base, Foundation OSB, and Foundation UI Services

For more information about setting up the RKE2 Kubernetes cluster and deploying Foundation services, check with the GE project team.

### 3.2.3. Deployment Server

HMS deployment requires the following software components:

- Docker version 18.09.3 or later
- Connection to a Docker Trusted Registry or a similar supported API/v2 registry (for example, Nexus)

### 3.2.4. Clients

HMS clients require the following software configurations:

- Google Chrome x64 version 100 or later with an enabled JavaScript engine.


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# 4. Deploying the Hazard Management and Conflict Manager

This section describes the steps required before deploying the Hazard Management and Conflict Manager solution to the Kubernetes cluster.

> **Note**
>
> Refer to the Foundation documentation for more details on Helm-based and ArgoCD-Based deployment and maintenance of YAML files in Nexus, Git or Local repo.

The Hazard Management and Conflict Manager deployment includes the following steps:

- Configuring the Foundation components
- Configuring the Hazard Management components
- Configuring the Conflict Manager components
- Deploying the Foundation, Hazard Management and Conflict Manager components
- Configuring the ADMS Services

## 4.1. Configuring Foundation Base Components

The Hazard Management and Conflict Manager solution uses Foundation Base, which includes security components of Foundation. For more information about Foundation, consult your GE project team.

The Foundation Base services use cryptographic keys and certificates in these areas:

- The Ingress Controller to perform TLS termination.
- Public/Private key pair for UAA to sign access tokens.

These keys and certificates must be created prior to deploying Foundation Base.

## 4.2. Configuring Hazard Management and Conflict Manager Components

### 4.2.1. Preparing Local Setup Folders

A sample folder structure that is used for deployment is shown below. This structure can be used as a reference to create the folder structure.

```
└──── k8s-base-manifest.yaml
```


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```
└──── Foundation helm chart
└──── Hazard Management helm chart
└──── Strimzi Kafka helm chart
└──── Conflict Manager helm chart
└──── apps
   └──── custom-namespaces.yaml
   └──── foundation-base.yaml
   └──── foundation-data.yaml
   └──── foundation-osb.yaml
   └──── foundation-ui-server.yaml
   └──── hazard-management-kafka.yaml
   └──── hms-data.yaml
   └──── hms.yaml
   └──── conflict-manager.yaml
└──── apps-config
   └──── hms-values.yaml
   └──── strimzi-kafka-values.yaml
   └──── conflict-manager-values.yaml
```

### 4.2.2. Generating Certificates for SSL/TLS Termination with ADMS

For production, it is recommended to use certificates issued by a certificate authority. For testing and demo purposes, self-signed certificates can be used.

For HMS, the following certification files are required:

- Root certificate authority certificate. This certificate is used to issue and sign certificates. (For example, file name `rootCA.crt`; CN HMS Root CA)
- Ingress controller certificate and key. These files are used for establishing connections with Active Directory. (For example, file name `ingress-local-tls.crt` and `ingress-local-tls.key`; ingress.local)
- ADMS server CRT certificate file signed by the Root certificate authority certificate. This file is used for secure communication from the Kubernetes cluster to ADMS interface servers. (For example, file name `adms-server.crt`; CN ADMS; O GE).

  The certificate Subject Alternative Name must include all DNS host names and IPs referenced in the HMS helm chart to enable the certificate for all ADMS IPs and host names. For example:


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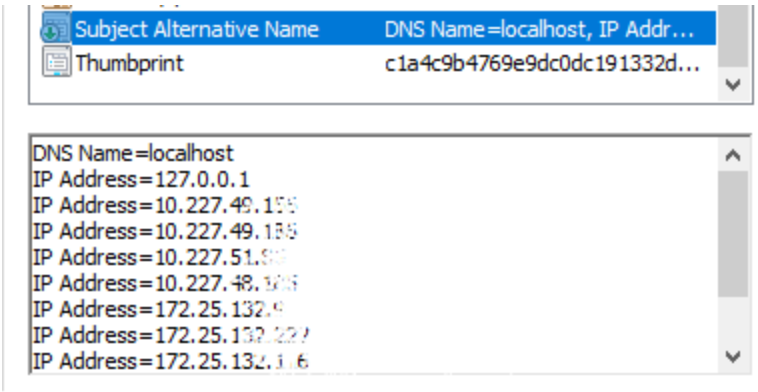


- ADMS server PFX file with a certificate (CRT file) and the associated private key. This file is used for certificate hosting by ADMS interface servers. (For example, file name `adms-server.pfx`; CN ADMS; O GE). The installed certificate has the same properties as the CRT file used in the Kubernetes cluster.

The example below shows how to generate OpenSSL certificates using the openssl tool. This is a standard tool for most Linux distributions. For more information, see https://www.openssl.org/source/.

**To generate self-signed certificates:**

1. Generate a root certificate authority certificate and key:

```
openssl genrsa -out rootCA.key 2048
openssl req -x509 -new -key rootCA.key -days 10000 -out rootCA.crt
```

2. Generate a certificate and key for Ingress Controller:

> **Note**
>
> It is recommended to set CN to the actual ingress controller DNS host name (web address you access when connecting to a secure site).

```
openssl genrsa -out ingress-local-tls.key 2048
openssl req -new -key ingress-local-tls.key -out ingress-local-tls.csr
openssl x509 -req -in ingress-local-tls.csr -CA rootCA.crt -CAkey rootCA.key -CAcreateserial -out
ingress-local-tls.crt -days 5000
```

3. Generate a certificate and key for ADMS servers using the openssl tool:
   - Create the csr.conf file with the IPs or DNS names of the ADMS machines. All DNS names can be listed in the file as DNS.1, DNS.2, DMS.N:

```
[v3_ext]
subjectAltName = @alt_names
[alt_names]
DNS.1 = localhost
```

```
DNS.2 = admshost1
DNS.3 = admshost2
DNS.4 = admshost3
DNS.5 = admshost4
IP.1 = 10.227.49.11
IP.2 = 10.227.49.12
IP.3 = 10.227.49.13
IP.4 = 10.227.49.14
```

- Execute the following commands using the Apache openssl tool supplied with the ADMS UX Prerequisites kit, where NetViewer Adapter uses O=GE value in <CertificateSubject>GE</CertificateSubject>.

```
"C:\Eterra\Distribution\PrerequisitesServices\Apache\bin\openssl.exe" genrsa -out adms-server.key 2048
```

This generates the ADMS Server key file, adms-server.key.

```
"C:\Eterra\Distribution\PrerequisitesServices\Apache\bin\openssl.exe" req -new -key adms-server.key -config "C:\Eterra\Distribution\PrerequisitesServices\Apache\conf\openssl.cnf" -out adms-server.csr -subj "/C=US/ST=WA/L=Bothell/O=GE/OU=Digital/CN=ADMS Server"
```

This generates the ADMS Server certificate signing request file, adms-server.csr, using adms-server.key.

```
"C:\Eterra\Distribution\PrerequisitesServices\Apache\bin\openssl.exe" x509 -req -in adms-server.csr -CA rootCA.crt -CAkey rootCA.key -CAcreateserial -out adms-server.crt -extensions v3_ext -extfile csr.conf -days 3650
```

This generates the ADMS Server certificate file, adms-server.crt, using csr.conf, adms-server.csr, rootCA.crt, and rootCA.key. Copy the content of `adms.crt` file in hms-values.yaml and conflict-manager-values.yaml files.

```
"C:\Eterra\Distribution\PrerequisitesServices\Apache\bin\openssl.exe" pkcs12 -export -out adms-server.pfx -inkey adms-server.key -in adms-server.crt
```

This generates the ADMS Server PFX file, which includes a certificate and key for Windows certificate installation or import. During the certificate installation, you can either set an empty password by pressing Enter to disable it or use a non-empty password to enable it.

> **Note**
>
> It is recommended that you set CN to the actual ADMS DNS host name (web address you access when connecting to a secure site).


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## 4.2.3. Setting Up Firewall Rules

Open the firewall ports as shown in the table below to enable communication among the Kubernetes cluster, ADMS servers, LDAP/AD Server.

| Source | Destination | Port | Protocol |
|---|---|---|---|
| HMS External Gateway (Kubernetes Cluster) | OMS Interface Server | 8812 | HTTPS |
| HMS External Gateway (Kubernetes Cluster) | NetViewer Adapter (ADMS) | 24468 | HTTPS |
| HMS External Gateway (Kubernetes Cluster) | DMS API Server | 50058 | HTTPS |
| HMS External Gateway (Kubernetes Cluster) | DMS API Server | 8801 | HTTPS |
| Kubernetes Cluster | LDAP/AD Server | Project-specific | LDAPS |
| ADMS Interface Servers<br>• DMS API Server<br>• OMS Interface Server<br>• DMS Interface Server<br>• NetViewer Adapter<br>• Hazard Management DS Service | Kubernetes Cluster | 443 | HTTPS |

> **Note**
>
> Port numbers and protocols can be configured in configuration files for the associated applications.

## 4.2.4. Hazard Management and Conflict Manager: Required Permissions and Roles

To use the **Hazard Management** application and **Conflict Manager** APIs, specific permissions and roles must be configured in the **Active Directory**.

### 4.2.4.1. Permissions

The following permissions control access to Hazard Management and Conflict Manager functionalities:

```
# Hazard Management
- permissionName: hms.write.writeAll
  description: Create, Edit and change the status of Hazard Events.
- permissionName: hms.read.readAll
```


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```
  description: Read Hazard Events
- permissionName: hms.write.edit
  description: Create and edit Hazard Events
# Conflict Manager
- permissionName: conflictManager.read.readAll
  description: Read conflict event data from the Conflict Manager service
- permissionName: conflictManager.write.writeAll
  description: Update Conflict Manager service configuration
```

### 4.2.4.2. Roles

The roles below define the aggregation of permissions for different user groups:

```
- name: HMS Administrator
  permissions: [
        'hms.read.readAll',
        'hms.write.writeAll',
        'hms.write.edit'
  ]

- name: HMS Editor
  permissions: [
        'hms.read.readAll',
        'hms.write.edit'
  ]

- name: HMS Viewer
  permissions: [
        'hms.read.readAll'
  ]

- name: ConflictManager Administrator
  permissions: [
    'conflictManager.read.readAll',
    'conflictManager.write.writeAll'
  ]

- name: ConflictManager Viewer
  permissions: [
    'conflictManager.read.readAll'
  ]
```

### 4.2.4.3. Usage Notes

Permissions ensure fine-grained control over the Hazard Management application and Conflict Manager API's. The application currently supports the following permissions:

- Users with the `hms.write.writeAll` permissions can create, modify and update status and delete Hazard Events.


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- Users with the `hms.write.edit` permissions can create, modify Hazard Events.
- Users with the `hms.read.readAll` permission can read Hazard Management Events.
- Users with the `conflictManager.read.readAll` permission can read conflict event data from the Conflict Manager service
- Users with the `conflictManager.write.writeAll` permission can modify the Conflict Manager configuration.

### 4.2.4.4. Default Role Assignments in Foundation Deployment

In the default **foundation deployment**, the following user groups are assigned Hazard Management roles:

1. **Supervisors**
   - Assigned Roles: `HMS Administrator`, `ConflictManager Administrator`
   - Privileges: Users in this group can create, modify and delete the Hazard Events and access Conflict Manager APIs.
2. **Operators**
   - Assigned Roles: `HMS Editor`, `ConflictManager Administrator`
   - Privileges: Users in this group can create, modify and delete the Hazard Events.
3. **Readonly Viewers**
   - Assigned Roles: `HMS Viewer`, `ConflictManager Viewer`
   - Privileges: Users in this group can view the Hazard Events and access Conflict Manager APIs.
4. **Service Account Group 5**
   - Assigned Roles: `HMS Administrator`, `ConflictManager Administrator`
   - Description: This group represents client authority defined in the UAA client. It enables the Hazard Management and Conflict Manager Clients to connect to the Kubernetes cluster and access Hazard Management, Conflict Manager APIs.

```
usergroups:
  - name: Operators
    roles: [
       'Operator',
       'Session Viewer',
       'Dashboard Viewer',
       'HMS Editor',
       'ConflictManager Administrator']
    ]

  - name: Supervisors
    roles: [
      'Supervisor',
```


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```
      'Kibana Viewer',
      'Grafana Viewer',
      'Prometheus Viewer',
      'Alertmanager Viewer',
      'Kiali Viewer',
      'Jaeger Viewer',
      'Policy Report Viewer',
      'Dashboard Administrator',
      'User Details Viewer',
      'HMS Administrator',
      'ConflictManager Administrator']
    ]

  - name: Readonly Viewers
    roles: [
      'Readonly Viewer',
      'HMS Viewer',
      'ConflictManager Viewer']

  - name: Service Account Group 5
    roles: [
      'Tile Service Consumer',
      'AORs Manager Sessions Consumer',
      'ES Adapter Service Consumer',
      'Preferences Service Consumer',
      'HMS Administrator',
      'ConflictManager Administrator']
```

This configuration ensures that users and service accounts have the appropriate access levels to manage Hazard Management and Conflict Manager effectively.

## 4.3. Preparing for Hazard Management and Conflict Manager Deployment

This section outlines the Helm files and configurations required to deploy the RKE2 Kubernetes cluster, Foundation services, and the Hazard Management (HMS) and Conflict Manager components.

### 4.3.1. Steps to Prepare YAML Files for HMS Deployment

1. **Prepare the Application Files**:

   Create the application and configuration files necessary for deploying the Foundation services and HMS.

2. **Configure IP Addresses**:

   Update the IP addresses to match your current setup.


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3. **Specify Docker Registry and GitHub**:
   - Provide the address of the Docker Registry and GitHub, such as GE's Docker Trusted Registry (DTR) or a Nexus registry on the deployment server.
   - If using a Nexus Docker Registry on a deployment server, ensure the `image` paths in the Docker configuration are updated to include the Nexus host and port.
4. **Set Deployment Parameters**:

   Configure deployment settings, including paths to the Docker registry, container names, and the components to be deployed.
5. **Configure Storage**:

   Update the `storageClass` properties with the actual names of the Local Persistent Volume storage.

## 4.3.2. Kubernetes cluster deployment

The RKE2 Kubernetes cluster deployment is defined using the k8s-base-manifest.yaml file.

1. Specify the address of the Docker Registry that contains YAML file (GE's DTR or a Nexus registry on a deployment server).
2. When deploying from a Nexus Docker Registry on a deployment server, update the docker "image" paths to use the Nexus host and port.
3. Configure the deployment settings, such as paths to the Docker registry, container names, and components to be deployed.

### 4.3.2.1. k8s-base-manifest.yaml

```
id: k8s-base-stack-{{ env['BRANCH_NAME'] }}
api_version: '5.3'
targets:
  - id: AWS
    products:
      - id: k8s-infra
        pdi:
          image: gart.software.gevernova.com/foundation-docker/k8s-infra-pdi
          tag: 'latest'
        local_dir_mount_path: './local/'
        param_groups:
          - id: environment
            params:
              #TF_VAR_operating_system: 'CENTOS7'
              TF_VAR_operating_system: 'RHEL8'
              TF_VAR_number_of_masters: '1'
              TF_VAR_number_of_agents: '3'
              TF_VAR_instance_size: 't3a.2xlarge'
```


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```
    - id: k8s
      pdi:
        image: gart.software.gevernova.com/foundation-docker/pdi-k8s
        tag: "{{ env['PDI_K8S_TAG'] }}"
      local_dir_mount_path: './local/'
      param_groups:
        - id: environment
          params:
            INVENTORY_FILE_PATH: '/data/local/k8s_inventory.yaml'
            # ANSIBLE_RUN_ARGS: '-vvvv'
            SONOBUOY_MODE: 'quick'    # modes: https://sonobuoy.io/docs/main/e2eplugin/
        - id: k8s_all_group_vars
          params:
            ansible_user: gecloud  # expected user for the image above
            ansible_ssh_private_key_file: '/data/local/ssh.pem'
            rke2_ingress_nginx: false

            # container_selinux_version: "2.107"
            # audit: true

            # Enable CIS profile for adding Istio CNI (istioCniEnable: true) in Foundation
            rke2_cis_profile: true
            selinux: true
```

## 4.3.3. Namespaces for Hazard Management and Conflict Manager.

The Hazard Management and Conflict Manager components are deployed in separate namespaces. Hazard Management resources are deployed in the `hms` namespace, while Conflict Manager resources are deployed in the `conflict-manager` namespace. These namespaces are specified in the `custom-namespaces.yaml` file.

### 4.3.3.1. custom-namespaces.yaml

```
namespaces:
 - name: foundation-cluster-monitoring
   istio_injected: true
 - name: foundation-cluster-operators
   istio_injected: true
 - name: foundation-cluster-zerotrust
   istio_injected: true
 - name: kyverno
   istio_injected: false
 - name: foundation-cluster-tools
   istio_injected: true
 - name: foundation-env-default
   istio_injected: true
```


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```
  - name: foundation-env-default
    istio_injected: true
  - name: oms
    istio_injected: true
  - name: hms
    istio_injected: true
  - name: conflict-manager
    istio_injected: true
  - name: conjur
    istio_injected: true
```

## 4.3.4. Strimzi Kafka Configuration

This section outlines the configurable parameters that need to be updated in the `strimzi-kafka-values.yaml` file located in the `apps-config` folder.

The table's below lists the parameters that can be configured in each section

| Field | Default Value | Description |
|---|---|---|
| image.repository | strimzi/kafka | Kafka image repository. |
| image.tag | 0.42.0-kafka-3.7.1 | Kafka image tag. This parameter is for reference only and should not be changed. |
| ingressClass | nginx | Ingress class for Kafka. This parameter is for reference only and should not be changed. |
| runAsUser | 1000 | User ID to run Kafka pods. This parameter is for reference only and should not be changed. |
| runAsGroup | 1000 | Group ID to run Kafka pods. This parameter is for reference only and should not be changed. |
| hosts | • name: broker-0.ingress.local id: 0<br>• name: broker-1.ingress.local id: 1<br>• name: broker-2.ingress.local id: 2 | Hostnames and IDs for Kafka brokers. |
| bootstrapAddress | bootstrap.ingress.local | Address for Kafka bootstrap server. |
| deployKafkaCluster | true | Whether to deploy the Kafka cluster. |
| deployKafkaMonitoringResources | false | Whether to deploy monitoring resources for Kafka. |
| kafka.injectIstio | "false" | Whether to inject Istio sidecar into Kafka pods. This parameter is for reference only and should not be changed. |
| kafka.replicas | 3 | Number of Kafka broker replicas. |
| kafka.maxUnavailable | 2 | Configuration for PodDisruptionBudget maximum number of unavailable Kafka brokers. |


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| Field | Default Value | Description |
|---|---|---|
| kafka.externalType | ingress | Kafka external listener type. This parameter is for reference only and should not be changed. |
| kafka.jvm.heap.min | 1G | Minimum JVM heap size for Kafka. |
| kafka.jvm.heap.max | 1G | Maximum JVM heap size for Kafka. |
| kafka.jvm.parallelGCThreads | "8" | Number of parallel garbage collection threads for Kafka. |
| kafka.listeners.internal.plain.port | 9092 | Port for internal plain text communication. |
| kafka.listeners.internal.tls.enabled | true | Whether TLS is enabled for internal communication. |
| kafka.listeners.internal.tls.port | 9093 | Port for internal TLS communication. |
| kafka.storage.type | persistent-claim | Storage type for Kafka. |
| kafka.storage.size | 10Gi | Storage size for Kafka. |
| kafka.storage.class | local-path | Storage class for Kafka. |
| kafka.config.offsetTopicReplicationFactor | 3 | Replication factor for offset topic. |
| kafka.config.transactionStateLogReplicationFactor | 3 | Replication factor for transaction state log. |
| kafka.config.transactionStateLogMinIsr | 2 | Minimum in-sync replicas for transaction state log. |
| kafka.config.messageMaxBytes | 536870912 | Maximum message size in bytes. |
| kafka.config.autoCreateTopicsEnable | false | Whether to allow auto-creation of topics. |
| kafka.config.logRetentionMs | 21600000 | Log retention period in milliseconds. |
| kafka.config.socketRequestMaxBytes | 104857600 | Maximum socket request size in bytes. |
| kafka.terminationGracePeriodSeconds | 120 | Grace period for Kafka termination. |
| kafka.loggerLevel | INFO | Logging level for Kafka. |
| zookeeper.injectIstio | "false" | Whether to inject Istio sidecar into Zookeeper pods. This parameter is for reference only and should not be changed. |
| zookeeper.name | zookeeper-cluster | Name of the Zookeeper cluster. |
| zookeeper.replicas | 3 | Number of Zookeeper replicas. |
| zookeeper.maxUnavailable | 1 | Configuration for PodDisruptionBudget maximum number of unavailable Zookeeper nodes. |
| zookeeper.jvm.heap.min | 2G | Minimum JVM heap size for Zookeeper. |
| zookeeper.jvm.heap.max | 2G | Maximum JVM heap size for Zookeeper. |
| zookeeper.jvm.parallelGCThreads | "8" | Number of parallel garbage collection threads for Zookeeper. |
| zookeeper.storage.type | persistent-claim | Storage type for Zookeeper. |


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| Field | Default Value | Description |
|---|---|---|
| zookeeper.storage.size | 10Gi | Storage size for Zookeeper. |
| zookeeper.storage.class | local-path | Storage class for Zookeeper. |
| global.image.registry | gart.software.gevernova.com/virtual-docker | Docker image registry for global resources. |
| global.image.pullPolicy | IfNotPresent | Image pull policy. |
| global.imagePullSecrets | docker-registry | Secret for pulling images. |
| global.kafkaClusterName | hazard-management | Name of the Kafka cluster. This parameter is for reference only and should not be changed. |
| global.strimziKafkaCluster.namespace | foundation-env-default | Namespace for Strimzi Kafka cluster. This parameter is for reference only and should not be changed. |
| global.strimziKafkaCluster.serviceAccountName | hazard-management-kafka | Service account for the Kafka cluster. This parameter is for reference only and should not be changed. |
| global.hms.namespace | hms | Namespace for HMS resources. This parameter is for reference only and should not be changed. |
| global.zeroTrustNamespace | foundation-cluster-zerotrust | Namespace for foundation zero trust services. This parameter is for reference only and should not be changed. |
| global.jwksUri | http://sws-uaa.foundation-cluster-zerotrust.svc.cluster.local:8080/uaa/token_keys | URI for JWKS token keys. This parameter is for reference only and should not be changed. |
| global.clusterExternalUrl | https://ingress.local | External URL for the cluster. |
| global.istioPolicies.enabled | true | Whether Istio policies are enabled. This parameter is for reference only and should not be changed. |

### 4.3.4.1. strimzi-kafka-values.yaml

```
# Default values for kafka.

image:
  repository: strimzi/kafka
  tag: 0.42.0-kafka-3.7.1

ingressClass: nginx
runAsUser: 1000
runAsGroup: 1000
hosts:
  - name: broker-0.ingress.local
    id: 0
  - name: broker-1.ingress.local
    id: 1
  - name: broker-2.ingress.local
    id: 2
```


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```
bootstrapAddress: bootstrap.ingress.local

deployKafkaCluster: true
deployKafkaMonitoringResources: false

kafka:
  injectIstio: "false"
  replicas: 3
  maxUnavailable: 2
  externalType: ingress
  jvm:
    heap:
      min: 1G
      max: 1G
    parallelGCThreads: "8"
  listeners:
    internal:
      plain:
        port: 9092
      tls:
        enabled: true
        port: 9093
  storage:
    type: "persistent-claim"
    size: "10Gi"
    class: "local-path"
  config:
    offsetTopicReplicationFactor: 3
    transactionStateLogReplicationFactor: 3
    transactionStateLogMinIsr: 2
    messageMaxBytes: 536870912
    autoCreateTopicsEnable: false
    logRetentionMs: 21600000
    socketRequestMaxBytes: 104857600
  terminationGracePeriodSeconds: 120
  loggerLevel: "INFO"

zookeeper:
  injectIstio: "false"
  name: zookeeper-cluster
  replicas: 3
  maxUnavailable: 1
  jvm:
    heap:
      min: 2G
      max: 2G
    parallelGCThreads: "8"
  storage:
    type: "persistent-claim"
    size: "10Gi"
    class: "local-path"

global:
```


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```
  image:
    registry: gart.software.gevernova.com/virtual-docker
    pullPolicy: IfNotPresent
  imagePullSecrets:
    - name: docker-registry
  kafkaClusterName: hazard-management
  strimziKafkaCluster:
    namespace: foundation-env-default
    serviceAccountName: hazard-management-kafka
  hms:
    namespace: hms
  zeroTrustNamespace: foundation-cluster-zerotrust
  jwksUri: http://sws-uaa.foundation-cluster-zerotrust.svc.cluster.local:8080/uaa/token_keys
  clusterExternalUrl: https://ingress.local
  istioPolicies:
    enabled: true
```

## 4.3.5. Hazard Management Configuration

This section outlines the configurable parameters that need to be updated in the `hms-values.yaml` file located in the `apps-config` folder.

The table's below lists the parameters that can be configured in each section.

| Field | Default Value | Description |
|---|---|---|
| hazard-management-kafka-topics. kafka.topics.notifications UiFoundation.replicas | 3 | Number of replicas for the notifications UI foundation Kafka topic. |
| hazard-management-kafka-topics. kafka.topics.notifications UiFoundation.partitions | 1 | Number of partitions for the notifications UI foundation Kafka topic. |
| hms-external-gateway. externalGateway.image | gart.software.gevernova.com /foundation-docker/http-gateway:1.9.8 | Image for the external gateway service. This parameter is for reference only and should not be changed. |
| hms-external-gateway. admsExternalGateway.useIns ecureTrustManager | false | Whether to use an insecure trust manager for ADMS external gateway. |
| hms-external-gateway. admsExternalGateway. loggingLevel | ERROR | Logging level for the ADMS external gateway. |
| hms-external-gateway. admsExternalGateway. replicas | 1 | Number of replicas for the ADMS external gateway. |
| hms-external-gateway. oisExternalGatewayRoute. endpoints[0] | http://{WIN_SERVER}:8802 | Endpoint for OIS external gateway route. Replace {WIN_SERVER} with the Server IP where OMS Interface Server is running. |


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| Field | Default Value | Description |
|---|---|---|
| hms-external-gateway.disExternalGatewayRoute.endpoints[0] | http://{WIN_SERVER}:8801 | Endpoint for DIS external gateway route. Replace {WIN_SERVER} with the Server IP where DMS Interface Server is running. |
| hms-external-gateway.nvaMainExternalGatewayRoute.endpoints[0] | http://{WIN_SERVER}:24468 | Endpoint for NVA Main external gateway route. Replace {WIN_SERVER} with the Server IP where NetViewer Adapter is running. |
| hms-external-gateway.dmsApiExternalGatewayRoute.endpoints[0] | http://{WIN_SERVER}:50058 | Endpoint for DMS API external gateway route. Replace {WIN_SERVER} with the Server IP where DMS API Server is running. |
| hms-external-gateway.admsCertSecret.name | adms-certificate-secret | Name of the ADMS certificate secret. |
| hms-external-gateway.admsCertSecret.cert | `<certificate-content>` | ADMS certificate content. |
| pref-init-hms.prefServiceUrl | http://preferences-service.foundation-env-default | URL for the preferences service. This parameter is for reference only and should not be changed. |
| postgres-schema-creator.image.repository | spilo-cdp | Repository for the Postgres schema creator image. This parameter is for reference only and should not be changed. |
| postgres-schema-creator.image.tag | 14-2.1-p6-build.7 | Tag for the Postgres schema creator image. This parameter is for reference only and should not be changed. |
| postgres-schema-creator.image.pullPolicy | IfNotPresent | Image pull policy for the Postgres schema creator. This parameter is for reference only and should not be changed. |
| hazard-management.image.repository | hazard-management | Repository for the hazard management image. This parameter is for reference only and should not be changed. |
| hazard-management.image.tag | 16.16.2-rc.1 | Tag for the hazard management image. This parameter is for reference only and should not be changed. |
| hazard-management.config.hmsEventTimer.heartBeat | "2" | Heartbeat interval for HMS events. |
| hazard-management.config.topologyChangeEventTimer.heartBeat | "3600" | Heartbeat interval for topology change events. |
| hazard-management.config.topologyChangeEventTimer.enable | "true" | Whether topology change events are enabled. |
| hazard-management.config.hazardManagementApp.userId | {8599AE31-2409-4875-9ECF-0698FAE23EDA} | User ID for the hazard management app. This parameter is for reference only and should not be changed. |
| hazard-management.config.aspnetCore.environment | Development | ASP.NET Core environment. |
| hazard-management.config.overrideAppSettings | true | If set to `true`, the ConfigMap referencing the `appsettings.json` file must be created beforehand by deploying the `hms-data` Helm chart. If set to `false`, the default `appsettings.json` file will be used. |
| hazard-management-ds.image.repository | hazard-management-ds | Repository for the hazard-management-ds image. This parameter is for reference only and should not be changed. |


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| Field | Default Value | Description |
|---|---|---|
| hazard-management-ds.image.tag | 16.16.2-rc.1 | Tag for the hazard-management-ds image. This parameter is for reference only and should not be changed. |
| hazard-management-ds.config.persistence.mode | NonArchive | Persistence mode for hazard management DS. The supported values are 'NonArchive' or 'Archive'. The default value is 'NonArchive' and should not be changed for Kubernetes deployment. |
| hazard-management-ds.config.persistence.db | Postgres | Database used by hazard management DS. This parameter is for reference only and should not be changed. |
| hazard-management-ds.config.overrideAppSettings | true | If set to `true`, the ConfigMap referencing the `appsettings.json` file must be created beforehand by deploying the `hms-data` Helm chart. If set to `false`, the default `appsettings.json` file will be used. |
| hazard-management-ui.image.repository | hazard-management-ui | Repository for the hazard-management-ui image. This parameter is for reference only and should not be changed. |
| hazard-management-ui.image.tag | 16.16.2-rc.4 | Tag for the hazard-management-ui image. This parameter is for reference only and should not be changed. |
| global.image.registry | gart.software.gevernova.com/virtual-docker | Registry for global images. |
| global.image.pullPolicy | Always | Image pull policy for global images. |
| global.hookUtilImage.repository | foundation-hook-utils | Repository for the hook utility image. |
| global.hookUtilImage.tag | 1.1.3-build.9 | Tag for the hook utility image. |
| global.hookUtilImage.pullPolicy | IfNotPresent | Image pull policy for the hook utility image. |
| global.imagePullSecrets[0].name | docker-registry | Secret name for pulling images. |
| global.zeroTrustNamespace | foundation-cluster-zerotrust | Namespace for zero-trust resources. |
| global.foundationEnvNamespace | foundation-env-default | Namespace for the foundation environment. |
| global.kafkaNamespace | foundation-env-default | Namespace for Kafka resources. |
| global.hmsNamespace | hms | Namespace for HMS resources. |
| global.clusterExternalUrl | http://ingress.local | External URL for the cluster. |
| global.fqdnSuffix | .svc.cluster.local | Suffix for fully qualified domain names. |
| global.jwksUri | http://sws-uaa.foundation-cluster-zerotrust.svc.cluster.local:8080/uaa/token_keys | URI for JSON Web Key Set (JWKS) tokens. |
| global.zitadel.enabled | false | Whether Zitadel integration is enabled. Zitadel is currently not supported, so this parameter should be set to `false`. |
| global.zitadel.organisationName | foundation | Name of the Zitadel organization. |
| global.postgres.servicePort | 5432 | Service port for PostgreSQL. |


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| Field | Default Value | Description |
|---|---|---|
| global.postgres.serviceHost | postgres-db-pg-cluster.foundation-cluster-zerotrust.svc.cluster.local | Hostname for PostgreSQL service. |
| global.postgres.hmsCreateSecret | true | Whether to create an HMS secret for PostgreSQL. |
| global.postgres.hmsSecretName | hms-db-credentials-secret | Name of the HMS credentials secret. |
| global.postgres.hmsUsername | hms_user | Username for the HMS database. |
| global.postgres.pgSecretName | postgres-db-pg-cluster-postgres-secret | Secret name for PostgreSQL credentials. |
| global.postgres.pgPoolSecretName | postgres-db-pg-cluster-pgpool-creds | Secret name for PostgreSQL pool credentials. |
| global.generatePasswordSecret | true | Indicates whether a UAA client secret should be generated. The UAA client is used by the Hazard Management Clients to access the Hazard Management APIs. |
| global.hazardManagement.client.authorities[0] | service.account.group5 | Authority for hazard management client. This parameter is for reference only and should not be changed. |
| global.dmsApiServer.endpoint | http://hms-external-gateway/dms/ | Endpoint for the DMS API server. This parameter is for reference only and should not be changed. |
| global.dmsApiServer.client.authorities[0] | service.account.group5 | Authority for DMS API server client. This parameter is for reference only and should not be changed. |
| global.dmsInterfaceServer.endpoint | http://hms-external-gateway/dis/ | Endpoint for the DMS interface server. This parameter is for reference only and should not be changed. |
| global.dmsInterfaceServer.client.authorities[0] | service.account.group5 | Authority for DMS interface server client. This parameter is for reference only and should not be changed. |
| global.netViewAdapter.client.authorities[0] | service.account.group5 | Authority for NetView adapter client. This parameter is for reference only and should not be changed. |
| global.omsInterfaceServer.endpoint | http://hms-external-gateway/ois/ | Endpoint for the OMS interface server. This parameter is for reference only and should not be changed. |
| global.omsInterfaceServer.client.authorities[0] | service.account.group5 | Authority for OMS interface server client. This parameter is for reference only and should not be changed. |
| global.conflictManager.endpoint | http://conflict-manager-service.conflict-manager.svc.cluster.local/ | Endpoint for the Conflict Manager service. This parameter is for reference only and should not be changed. |
| global.hazardManagementDbServer.endpoint | http://hazard-management-ds/ | Endpoint for the Hazard Management database server. This parameter is for reference only and should not be changed. |
| global.notificationServer.endpoint | http://notification-service.foundation-env-default.svc.cluster.local/ | Endpoint for the notification server. This parameter is for reference only and should not be changed. |

### 4.3.5.1. hms-values.yaml

```
# Default values for breaker summary deployment.
```


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```
# This is a YAML-formatted file.
# Declare variables to be passed into your templates.

hazard-management-kafka-topics:
  kafka:
    topics:
      notificationsUiFoundation:
        replicas: 3
        partitions: 1

hms-external-gateway:
  externalGateway:
    image: gart.software.gevernova.com/foundation-docker/http-gateway:1.9.8
  admsExternalGateway:
    useInsecureTrustManager: false
    loggingLevel: ERROR
    replicas: 1
    maxHttpRequestHeaderSize: "1MB"
  oisExternalGatewayRoute:
    endpoints:
      - http://{WIN_SERVER}:8802
  disExternalGatewayRoute:
    endpoints:
      - http://{WIN_SERVER}:8801
  nvaMainExternalGatewayRoute:
    endpoints:
      - http://{WIN_SERVER}:24468
  dmsApiExternalGatewayRoute:
    endpoints:
      - http://{WIN_SERVER}:50058
#  conflictManagerExternalGatewayRoute:
#    endpoints:
#      - http://{WIN_SERVER}:5052
  admsCertSecret:
    name: adms-certificate-secret
    cert: |
      -----BEGIN CERTIFICATE-----
      MIIDozCCAougAwIBAgIUU3KvyEumb+TRqvDf5D9kxCipiKwwDQYJKoZIhvcNAQEL
      BQAwYTELMAkGA1UEBhMCQVUxEzARBgNVBAgMClNvbWUtU3RhdGUxEzARBgNVBAoM
      CkdFIERpZ2l0YWwxKDAmBgNVBAMMH0RFMi4wIEZvdW5kYXRpb24gQmFzZSAtIFJv
      b3QgQ0EwHhcNMjEwNDIxMDY0OTQ0WhcNMzEwNDE5MDY0OTQ0WjBhMQswCQYDVQQG
      EwJBVTETMBEGA1UECAwKU29tZS1TdGF0ZTETMBEGA1UECgwKR0UgRGlnaXRhbDEo
      MCYGA1UEAwwfREUyLjAgRm91bmRhdGlvbiBCYXNlIC0gUm9vdCBDQTCCASIwDQYJ
      KoZIhvcNAQEBBQADggEPADCCAQoCggEBAOrcZK4NBOQuunFJ8Di5bhL+PDwP6fDd
      Zdx06w0tOvguq2KagDFJjHa0NyChLHLJQdKHQ+ga2kY4Wqxt1wOlhCvC0gxVYQwV
      IHVinzbFD1/KxUGy6n2AIxhcGTLL+pbYL6e54lsl+T+orN76YYCt72o6cZlpwaLh
      ODfTfLX4sP/ljLYJmIw/X0dR85NoDiAVtm2S9pcdlO/ivorJnlq39LJ+LgjgIkG/
      FRJpMil/yqOTwnizq/M9kaM8HClNUYRzM+m8eIIXCM7hJrpO6OGWl3ken6BW0QXf
      NPPw9CWpRxTHAZcWsInvoCY+MwD3g9mQ6xHvzbXjgYIoVNMwEqpHydkCAwEAAaNT
      MFEwHQYDVR0OBBYEFAaNoDyDSLFFbNxaBJ7PDKFW0rQ0MB8GA1UdIwQYMBaAFAaN
      oDyDSLFFbNxaBJ7PDKFW0rQ0MA8GA1UdEwEB/wQFMAMBAf8wDQYJKoZIhvcNAQEL
      BQADggEBAEMA0F5u3aiHyfNDJDV3L2me2e1eKTqojr7oMHDNGdfJNqtJ7A1PMwor
      RutqYY6Jwe3j08fuIyF9EezSO305LgLZS683jkN3KidwSXlNpDE4aMo1XlTxA+12
```


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```
      LwbXXLTORMOKP1FQH1kRgaTKQKnqCbW1Jet/AyR9DIiPjSjPwVJJm3Pm1ztg4w0e
      jfkDhMU1QiqSQ7aviaC0Kiy8VYiDQLo0GitrFo+C+uaQKXwCAhRIRBZbLnBgFP6Q
      SWtS9o4M0f1tiqCSUEhT8wTYf2pEqPDMtEx5res8sdg+P45cIphFpB7pGojIDzKs
      uVZAiK58024KW49o74I+QrMbWqeH0mc=
      -----END CERTIFICATE-----

postgres-schema-creator:
  image:
    repository: spilo-cdp
    tag: 14-2.1-p6-build.7
    pullPolicy: IfNotPresent

hazard-management:
  image:
    repository: hazard-management
    tag: 16.18.0-rc.7
  config:
    hmsEventTimer:
      heartBeat: "2"
    topologyChangeEventTimer:
      heartBeat: "3600"
      enable: "true"
    hazardManagementApp:
      userId: "{8599AE31-2409-4875-9ECF-0698FAE23EDA}"
    aspnetCore:
      environment: Development
    overrideAppSettings: true

hazard-management-ds:
  image:
    repository: hazard-management-ds
    tag: 16.18.0-rc.7
  config:
    aspnetCore:
      environment: Development
    persistence:
      mode: "NonArchive"
      db: "Postgres"
    overrideAppSettings: true

hazard-management-ui:
  image:
    repository: hazard-management-ui
    tag: 16.18.0-rc.6

global:
  image:
    registry: gart.software.gevernova.com/virtual-docker
    pullPolicy: Always
  hookUtilImage:
    repository: foundation-hook-utils
    tag: 1.1.3-build.9
    pullPolicy: IfNotPresent
```




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```
  imagePullSecrets:
    - name: docker-registry
  zeroTrustNamespace: foundation-cluster-zerotrust
  foundationEnvNamespace: foundation-env-default
  kafkaNamespace: foundation-env-default
  hmsNamespace: hms
  clusterExternalUrl: http://ingress.local
  fqdnSuffix: .svc.cluster.local
  jwksUri: http://sws-uaa.foundation-cluster-zerotrust.svc.cluster.local:8080/uaa/token_keys
  zitadel:
    enabled: true
    organisationName: "foundation"
  postgres:
    servicePort: "5432"
    serviceHost: postgres-db-pg-cluster.foundation-cluster-zerotrust.svc.cluster.local
    hmsCreateSecret: true
    hmsSecretName: hms-db-credentials-secret
    hmsUsername: hms_user
    pgSecretName: postgres-db-pg-cluster-postgres-secret
    pgPoolSecretName: postgres-db-pg-cluster-pgpool-creds
  generatePasswordSecret: true
  hazardManagement:
    client:
      authorities: # Needs to match etd group defined in roles-config YAML
        - service.account.group5
  dmsApiServer:
    endpoint: http://hms-external-gateway/dms/
    client:
      authorities: # Needs to match etd group defined in roles-config YAML
        - service.account.group5
  dmsInterfaceServer:
    endpoint: http://hms-external-gateway/dis/
    client:
      authorities: # Needs to match etd group defined in roles-config YAML
        - service.account.group5
  netViewAdapter:
    client:
      authorities: # Needs to match etd group defined in roles-config YAML
        - service.account.group5
  omsInterfaceServer:
    endpoint: http://hms-external-gateway/ois/
    client:
      authorities: # Needs to match etd group defined in roles-config YAML
        - service.account.group5
  conflictManager:
    endpoint: http://conflict-manager-service.conflict-manager.svc.cluster.local/
  hazardManagementDbServer:
    endpoint: http://hazard-management-ds/
  notificationServer:
    endpoint: http://notification-service.foundation-env-default.svc.cluster.local/
```


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## 4.3.6. Creating secrets for Hazard Management

The following secrets are required for Hazard Management:

- Database Credentials
- UAA Client Secret

The user can choose to create these secrets manually or let the deployment process create them. Note that all fields in a secret are base64 encoded.

### 4.3.6.1. Database Credentials

- The database credentials are stored in the secret `hms-db-credentials-secret` in the `hms` namespace.
- The flag `global.postgres.hmsCreateSecret` in the `hms-values.yaml` must be set to `false` to create this secret manually.

Below is an example of the secret where both `username` and `password` is set to the base64-encoded value of the string `hms_user`:

```
apiVersion: v1
kind: Secret
metadata:
  name: hms-db-credentials-secret
  namespace: hms
  annotations:
    "helm.sh/hook": pre-install,pre-upgrade
    "helm.sh/hook-weight": "3"
type: Opaque
data:
  hms-postgres-db-username: aG1zX3VzZXI=
  hms-postgres-db-password: aG1zX3VzZXI=
```

### 4.3.6.2. UAA Client Secret

- The UAA Client secret is stored in the secret `hazard-management-client-secret` in the `hms` namespace.
- The flag `global.generatePasswordSecret` in the `hms-values.yaml` must be set to `false` to create this secret manually.

Below is an example of the secret for UAA Client with the secret value set to the base64-encoded value of the string `hazard-management-client-secret-pass`:

```
apiVersion: v1
kind: Secret
metadata:
  name: hazard-management-client-secret
```


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```
  namespace: etd
  annotations:
    "helm.sh/hook": pre-install,pre-upgrade
    "helm.sh/hook-weight": "3"
type: Opaque
data:
  secret: aGF6YXJkLW1hbmFnZW1lbnQtY2xpZW50LXNlY3JldC1wYXNz
```

## 4.3.7. Conflict Manager Configuration

This section outlines the configurable parameters that need to be updated in the `conflict-manager-values.yaml` file located in the `apps-config` folder.

The table's below lists the parameters that can be configured in each section.

| Field | Default Value | Description |
|---|---|---|
| conflict-manager-external-gateway. externalGateway.image | gart.software.gevernova.com /foundation-docker/http-gateway:1.9.8 | Image for the external gateway service for Conflict Manager. |
| conflict-manager-external-gateway. admsExternalGateway.name | conflict-manager-external-gateway | Name of the ADMS external gateway. This parameter is for reference only and should not be changed. |
| conflict-manager-external-gateway. admsExternalGateway. useInsecureTrustManager | false | Whether to use an insecure trust manager for ADMS external gateway. |
| conflict-manager-external-gateway. admsExternalGateway. loggingLevel | ERROR | Logging level for the ADMS external gateway. |
| conflict-manager-external-gateway. admsExternalGateway. replicas | 1 | Number of replicas for the ADMS external gateway. |
| conflict-manager-external-gateway. oisExternalGatewayRoute. endpoints[0] | http://{WIN_SERVER}:8802 | Endpoint for OMS Interface Server. Replace {WIN_SERVER} with the Server IP where OMS Interface Server is running. |
| conflict-manager-external-gateway. disExternalGatewayRoute. endpoints[0] | http://{WIN_SERVER}:8801 | Endpoint for DMS Interface Server. Replace {WIN_SERVER} with the Server IP where DMS Interface Server is running. |
| conflict-manager-external-gateway. dmsApiExternalGatewayRout e.endpoints[0] | http://{WIN_SERVER}:50058 | Endpoint for DMS API Server. Replace {WIN_SERVER} with the Server IP where DMS API Server is running. |
| conflict-manager-external-gateway. admsCertSecret.name | adms-certificate-secret | Name of the ADMS certificate secret. |


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| Field | Default Value | Description |
|---|---|---|
| conflict-manager-external-gateway.admsCertSecret.cert | `<certificate-content>` | Certificate content for the ADMS gateway. |
| conflict-manager-service. config.conflictManagerEventTimer. heartBeat | 500 ms | Heartbeat interval for conflict manager events. |
| conflict-manager-service. config.conflictManagerApp. userId | {9E91D772-2619-4085-B67C-46C9F6D791EC} | User ID for the Conflict Manager application. This parameter is for reference only and should not be changed. |
| conflict-manager-service. config.aspnetCore. environment | Development | ASP.NET Core environment for Conflict Manager. |
| global.image.registry | gart.software.gevernova.com /virtual-docker | Global image registry. |
| global.image.pullPolicy | Always | Global image pull policy. |
| global.imagePullSecrets[0].name | docker-registry | Name of the image pull secret. |
| global.zeroTrustNamespace | foundation-cluster-zerotrust | Namespace for zero-trust resources. This parameter is for reference only and should not be changed. |
| global.conflictManagerNamespace | conflict-manager | Namespace for Conflict Manager resources. This parameter is for reference only and should not be changed. |
| global.clusterExternalUrl | http://service.ingress.local | External URL for the cluster. |
| global.fqdnSuffix | .svc.cluster.local | Fully qualified domain name suffix. |
| global.jwksUri | http://sws-uaa.foundation-cluster-zerotrust.svc.cluster.local:8080/uaa/token_keys | URI for JSON Web Key Set (JWKS). This parameter is for reference only and should not be changed. |
| global.zitadel.enabled | false | Whether Zitadel integration is enabled. Zitadel is currently not supported, so this parameter should be set to `false`. |
| global.zitadel.organisationName | foundation | Name of the Zitadel organization. |
| global.generate PasswordSecret | true | Indicates whether a UAA client secret should be generated. The UAA client is used by the Conflict Manager Clients to access the Conflict Manager APIs. |
| global.conflictManager. client.authorities[0] | service.account.group5 | Authority for Conflict Manager client. This parameter is for reference only and should not be changed. |
| global.dmsApiServer. endpoint | http://conflict-manager-external-gateway/dms/ | Endpoint for the DMS API server. This parameter is for reference only and should not be changed. |
| global.dmsApiServer. client.authorities[0] | service.account.group5 | Authority for DMS API server client. This parameter is for reference only and should not be changed. |
| global.dmsInterfaceServer. endpoint | http://conflict-manager-external-gateway/dis/ | Endpoint for the DMS interface server. This parameter is for reference only and should not be changed. |
| global.dmsInterfaceServer. client.authorities[0] | service.account.group5 | Authority for DMS interface server client. This parameter is for reference only and should not be changed. |


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| Field | Default Value | Description |
|---|---|---|
| global.omsInterfaceServer.endpoint | http://conflict-manager-external-gateway/ois/ | Endpoint for the OMS interface server. This parameter is for reference only and should not be changed. |
| global.omsInterfaceServer.client.authorities[0] | service.account.group5 | Authority for OMS interface server client. This parameter is for reference only and should not be changed. |
| global.hazardManagementServer.endpoint | http://hazard-management.hms.svc.cluster.local/ | Endpoint for the Hazard Management server. This parameter is for reference only and should not be changed. |
| global.notificationServer.endpoint | http://notification-service.foundation-env-default.svc.cluster.local/ | Endpoint for the Notification server. This parameter is for reference only and should not be changed. |

## 4.3.7.1. conflict-manager-values.yaml

```
conflict-manager-external-gateway:
  externalGateway:
    image: gart.software.gevernova.com/foundation-docker/http-gateway:1.9.8

  admsExternalGateway:
    name: conflict-manager-external-gateway
    useInsecureTrustManager: false
    loggingLevel: ERROR
    replicas: 1

  oisExternalGatewayRoute:
    endpoints:
      - http://{WIN_SERVER}:8802

  disExternalGatewayRoute:
    endpoints:
      - http://{WIN_SERVER}:8801

  dmsApiExternalGatewayRoute:
    endpoints:
      - http://{WIN_SERVER}:50058

  admsCertSecret:
    name: adms-certificate-secret
    cert: |
      -----BEGIN CERTIFICATE-----
      MIIDozCCAougAwIBAgIUU3KvyEumb+TRqvDf5D9kxCipiKwwDQYJKoZIhvcNAQEL
      BQAwYTELMAkGA1UEBhMCQVUxEzARBgNVBAgMClNvbWUtU3RhdGUxEzARBgNVBAoM
      CkdFIERpZ2l0YWwxKDAmBgNVBAMMH0RFMi4wIEZvdW5kYXRpb24gQmFzZSAtIFJv
      b3QgQ0EwHhcNMjEwNDIxMDY0OTQ0WhcNMzEwNDE5MDY0OTQ0WjBhMQswCQYDVQQG
      EwJBVTETMBEGA1UECAwKU29tZS1TdGF0ZTETMBEGA1UECgwKR0UgRGlnaXRhbDEo
      MCYGA1UEAwwfREUyLjAgRm91bmRhdGlvbiBCYXNlIC0gUm9vdCBDQTCCASIwDQYJ
      KoZIhvcNAQEBBQADggEPADCCAQoCggEBAOrcZK4NBOQuunFJ8Di5bhL+PDwP6fDd
      Zdx06w0tOvguq2KagDFJjHa0NyChLHLJQdKHQ+ga2kY4Wqxt1wOlhCvC0gxVYQwV
      IHVinzbFD1/KxUGy6n2AIxhcGTLL+pbYL6e54lsl+T+orN76YYCt72o6cZlpwaLh
      ODfTfLX4sP/ljLYJmIw/X0dR85NoDiAVtm2S9pcdlO/ivorJnlq39LJ+LgjgIkG/
```


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```
      FRJpMil/yqOTwnizq/M9kaM8HClNUYRzM+m8eIIXCM7hJrpO6OGWl3ken6BW0QXf
      NPPw9CWpRxTHAZcWsInvoCY+MwD3g9mQ6xHvzbXjgYIoVNMwEqpHydkCAwEAAaNT
      MFEwHQYDVR0OBBYEFAaNoDyDSLFFbNxaBJ7PDKFW0rQ0MB8GA1UdIwQYMBaAFAaN
      oDyDSLFFbNxaBJ7PDKFW0rQ0MA8GA1UdEwEB/wQFMAMBAf8wDQYJKoZIhvcNAQEL
      BQADggEBAEMA0F5u3aiHyfNDJDV3L2me2e1eKTqojr7oMHDNGj0JNqtJ7A1PMwor
      RutqYY6Jwe3j08fuIyF9EezSO305LgLZS683jkN3KidwSXlNpDE4aMo1XlTxA+12
      LwbXXLTORMOKP1FQH1kRgaTKQKnqCbW1Jet/AyR9DIiPjSjPwVJJm3Pm1ztg4w0e
      jfkDhMU1QiqSQ7aviaC0Kiy8VYiDQLo0GitrFo+C+uaQKXwCAhRIRBZbLnBgFP6Q
      SWtS9o4M0f1tiqCSUEhT8wTYf2pEqPDMtEx5res8sdg+P45cIphFpB7pGojIDzKs
      uVZAiK58024KW49o74I+QrMbWqeH0mc=
      -----END CERTIFICATE-----

conflict-manager-service:
  config:
    conflictManagerEventTimer:
      heartBeat: "500"
    conflictManagerApp:
      userId: "{9E91D772-2619-4085-B67C-46C9F6D791EC}"
    aspnetCore:
      environment: Development

global:
  image:
    registry: gart.software.gevernova.com/virtual-docker
    pullPolicy: Always
  imagePullSecrets:
    - name: docker-registry
  zeroTrustNamespace: foundation-cluster-zerotrust
  conflictManagerNamespace: conflict-manager
  clusterExternalUrl: http://ingress.local
  fqdnSuffix: .svc.cluster.local
  jwksUri: http://sws-uaa.foundation-cluster-zerotrust.svc.cluster.local:8080/uaa/token_keys
  zitadel:
    enabled: true
    organisationName: "foundation"
  generatePasswordSecret: true
  # Conflict Manager Client configuration
  conflictManager:
    client:
      authorities: # Needs to match etd group defined in roles-config YAML
        - service.account.group5
  dmsApiServer:
    endpoint: http://conflict-manager-external-gateway/dms/
    client:
      authorities:
        - service.account.group5
  dmsInterfaceServer:
    endpoint: http://conflict-manager-external-gateway/dis/
    client:
      authorities:
        - service.account.group5
  omsInterfaceServer:
    endpoint: http://conflict-manager-external-gateway/ois/
```




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```
    client:
      authorities:
        - service.account.group5
  hazardManagementServer:
    endpoint: http://hazard-management.hms.svc.cluster.local/
  notificationServer:
    endpoint: http://notification-service.foundation-env-default.svc.cluster.local/
```

## 4.3.8. Hazard Event Types and Log Level Configuration

The default deployment of the Hazard Management Services (HMS) includes a set of predefined hazard event types. These event types are specified in the `appsettings.json` file, which holds the configuration settings for the Hazard Management Service and Hazard Management DS Service.

### 4.3.8.1. Default `appsettings.json` File Locations

In the provided tar file, the default `appsettings.json` files are located as follows:

- For the Hazard Management Service: `charts/hms/hazard-management/config/appsettings.json`
- For the Hazard Management DS Service: `charts/hms/hazard-management-ds/config/appsettings.json`

### 4.3.8.2. Using Default Settings

To use the default `appsettings.json` file for each service:

- Set `hazard-management.config.overrideAppSettings` to `false` in the `hms-values.yaml` file for the Hazard Management Service.
- Set `hazard-management-ds.config.overrideAppSettings` to `false` in the `hms-values.yaml` file for the Hazard Management DS Service.

### 4.3.8.3. Overriding `appsettings.json` with Helm Charts

A Helm chart, `hms-data`, is available to override the default `appsettings.json` files for both the Hazard Management and Hazard Management DS Services.

If `overrideAppSettings` is set to `true`, the following steps are required: . Deploy the `hms-data` Helm chart **before** deploying the Hazard Management and Hazard Management DS Helm charts. . The `hms-data` Helm chart creates two ConfigMaps: * `hms-app-settings-config` - Contains the overridden `appsettings.json` for the Hazard Management Service. * `hms-ds-app-settings-config` - Contains the overridden `appsettings.json` for the Hazard Management DS Service.

These ConfigMaps are essential for the successful deployment of the Hazard Management and Hazard Management DS Helm charts.


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## 4.3.8.4. Configuring Hazard Event Types

The following settings can be modified through `appsettings.json` of the Hazard Management Service:

- Hazard Event types
- Application logging level

> **Note**
>
> Only the Hazard Event Types and Application Log Levels are managed through the `appsettings.json` file. All other configurations are controlled by the values defined in the `hms-values.yaml` file.

### 4.3.8.4.1. Hazard Event Types

The `HazardTypes` section in the `appsettings.json` file defines the hazard event types that can be created in the HMS application. Each hazard event type has the following parameters:

1. **Name**
   - The full name of the hazard event type or category. It serves as a descriptive identifier used within the system to classify events.
2. **ShortName**
   - A short form or abbreviation of the hazard event type's name.
3. **Prefix**::
   - A string prefix added to identifiers related to this hazard event type. It ensures a consistent naming convention for resources, such as event IDs.
4. **PolygonColor**
   - Polygon colors are configured for below set of attributes based on hazard types.
     - ActiveStrokeColor
     - ActiveFillColor
     - PendingStrokeColor
     - PendingFillColor
     - WatchStrokeColor
     - WatchFillColor
     - PolygonHighlighter
5. **Restrictions**
   - A list of numeric restriction codes defining specific rules or constraints applicable to this hazard event type. These codes correspond to predefined restrictions in the system.


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#### 4.3.8.4.2. Restrictions

Below are the restrictions that can be applied for each event

Chose only one among restrictions 1, 2, 3 for an event

- 1 - Disable reclosing on recloser/breaker in zone
- 2 - Disable reclosing on reclosers in zone plus next upstream
- 3 - Disable reclosing on reclosers in zone plus all upstream
- 4 - Create a PLOS plan for system de-energization

Chose only one among restrictions 5, 6, 7 for an event

- 5 - Tag SCADA switches in zone as Do Not Operate
- 6 - Tag SCADA switches in zone plus next upstream as Do Not Operate
- 7 - Tag SCADA switches in zone plus all upstream as Do Not Operate
- 8 - Disable LVM application
- 9 - Disable FISR application
- 10 - Disable AFR application

Chose only one among restrictions 11, 12, 13 for an event

- 11 - Set protection in zone for fast trip point
- 12 - Set protection in zone plus next upstream for fast trip point
- 13 - Set protection in zone plus all upstream for fast trip point

Chose only one among restrictions 14, 15, 16 for an event

- 14 - Block load shed de-energization
- 15 - Block load shed re-energization
- 16 - Block load shed both de-energization and re-energization

Chose only one among restrictions 17, 18, 19 for an event

- 17 - Disable GroundTrip/breaker in zone
- 18 - Disable GroundTrip in zone plus next upstream
- 19 - Disable GroundTrip in zone plus all upstream

#### 4.3.8.4.3. Logging Level

The application log level for the Hazard Management Service and Hazard Management DS Service can be modified in the Logging section of the `appsettings.json` file.


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## 4.3.8.5. Hazard Management - appsettings.json

```
{
    "Logging": {
        "LogLevel": {
            "Default": "Debug",
            "Microsoft": "Warning"
        },
        "Debug": {
            "LogLevel": {
                "Default": "Trace",
                "Microsoft": "Warning"
            }
        },
        "Console": {
            "LogLevel": {
                "Default": "Trace",
                "Microsoft": "Warning"
            }
        }
    },
    "AllowedHosts": "*",
    "Kestrel": {
        "Endpoints": {
            "Http": {
                "Url": "http://*:30001"
            }
        }
    },
    "HazardManagementApp": {
        "UserId": "{8599AE31-2409-4875-9ECF-0698FAE23EDA}"
    },
    "DmsApi": {
        "Server": "http://localhost:50058/"
    },
    "DmsInterface": {
        "Server": "http://localhost:8801/"
    },
    "OmsInterface": {
        "Server": "http://localhost:8802/"
    },
    "HazardManagementDb": {
        "Server": "http://localhost:30002/"
    },
    "Notification": {
        "Server": "http://notification-service.foundation-env-default.svc.cluster.local/"
    },
    "RolesManager": {
        "Server": "http://authz-roles-manager.foundation-cluster-zerotrust.svc.cluster.local/"
    },
    "AorManager": {
        "Server": "http://authz-aors-manager.foundation-cluster-zerotrust.svc.cluster.local/"
    },
```


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```
    "ConflictManager": {
        "Server": "http://localhost:5052/"
    },
    "HazardTypes": [
        {
            "Name": "PSPS",
            "ShortName": "PSPS",
            "Prefix": "HZ-PSPS-",
            "InitialState": "Watch",
            "PolygonColor": {
                "ActiveStrokeColor": "#D97A14",
                "ActiveFillColor": "#D97A14",
                "PendingStrokeColor": "#7C899B",
                "PendingFillColor": "#7C899B",
                "WatchStrokeColor": "#DDE5B6",
                "WatchFillColor": "#DDE5B6"
            },
            "Restrictions": [
                2,
                4,
                6,
                8,
                9,
                10,
                12,
                16,
                18
            ]
        },
        {
            "Name": "Fire Weather Threat",
            "ShortName": "FWT",
            "Prefix": "HZ-FWT-",
            "InitialState": "Pending",
            "PolygonColor": {
                "ActiveStrokeColor": "#D97A14",
                "ActiveFillColor": "#D97A14",
                "PendingStrokeColor": "#7C899B",
                "PendingFillColor": "#7C899B",
                "WatchStrokeColor": "#DDE5B6",
                "WatchFillColor": "#DDE5B6"
            },
            "Restrictions": [
                2,
                8,
                9,
                10,
                12,
                18
            ]
        },
        {
            "Name": "Red Flag Warning",
```


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```
            "ShortName": "RFW",
            "Prefix": "HZ-RFW-",
            "InitialState": "Pending",
            "PolygonColor": {
                "ActiveStrokeColor": "#D97A14",
                "ActiveFillColor": "#D97A14",
                "PendingStrokeColor": "#7C899B",
                "PendingFillColor": "#7C899B",
                "WatchStrokeColor": "#DDE5B6",
                "WatchFillColor": "#DDE5B6"
            },
            "Restrictions": [
                2,
                8,
                9,
                10,
                12,
                18
            ]
        },
        {
            "Name": "Fire Climate Zone",
            "ShortName": "FCZ",
            "Prefix": "HZ-FCZ-",
            "InitialState": "Pending",
            "PolygonColor": {
                "ActiveStrokeColor": "#D97A14",
                "ActiveFillColor": "#D97A14",
                "PendingStrokeColor": "#7C899B",
                "PendingFillColor": "#7C899B",
                "WatchStrokeColor": "#DDE5B6",
                "WatchFillColor": "#DDE5B6"
            },
            "Restrictions": [
                2,
                8,
                9,
                10,
                12,
                18
            ]
        },
        {
            "Name": "Custom",
            "Prefix": "HZ-CST-",
            "InitialState": "Pending",
            "ShortName": "Custom",
            "PolygonColor": {
                "ActiveStrokeColor": "#D97A14",
                "ActiveFillColor": "#D97A14",
                "PendingStrokeColor": "#7C899B",
                "PendingFillColor": "#7C899B",
                "WatchStrokeColor": "#DDE5B6",
```


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```
                "WatchFillColor": "#DDE5B6",
                "PolygonHighlighter": "#1E90FF"
            },
            "Restrictions": []
        }
    ],
    "HmsEventTimer": {
        "Heartbeat": 2
    },
    "HmsAutoDeleteEventTimer": {
        "Heartbeat": 200
    },
    "TopologyChangeEventTimer": {
        "Heartbeat": 3600,
        "Enable": "true"
    },
    "LogLevelTimer": {
        "Heartbeat": 3600,
        "Enable": "true"
    },
    "AppVersion": "16.23.0",
    "HMSUser": "HMS",
    "AppConfig": {
        "timeZone": "",
        "dateTimeFormat": "yyyy-MM-dd HH:mm:ss"
    },
    "Permission": {
        "Admin": "hms.write.writeAll"
  }
}
```

## 4.3.8.6. Hazard Management DS - appsettings.json

```
{
  "Logging": {
    "LogLevel": {
      "Default": "Debug",
      "Microsoft": "Warning"
    },
    "Debug": {
      "LogLevel": {
        "Default": "Trace",
        "Microsoft": "Warning"
      }
    },
    "Console": {
      "LogLevel": {
        "Default": "Trace",
        "Microsoft": "Warning"
      }
    }
```


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```
  },
  "AllowedHosts": "*",
  "Kestrel": {
    "Endpoints": {
      "Http": {
        "Url": "http://*:30002"
      }
    }
  },
  "Persistence": {
    "Mode": "NonArchive",
    "DB": "Postgres",
    "ConnectionString":
"Host=localhost;Port=5432;Username=postgres;Password=postgres;Database=hazard_management"
  },
  "HazardManagementDs": {
    "Server": "http://localhost:30002/"
  },
  "HazardManagement": {
    "Server": "http://localhost:30001/"
  },
  "HmsEventTimer": {
    "Heartbeat": 2
  }
}
```

## 4.3.9. Foundation Base App Configuration

This section is applicable for ArgoCD based deployment. Specify the address of the Docker Registry and GitHub [GE's Docker Trusted Registry (DTR) or a Nexus registry on a deployment server] in the `foundation-base.yaml`.

In the tar file, the `foundation-base.yaml` file can be found at `apps`.

### 4.3.9.1. foundation-base.yaml

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: foundation-base
  namespace: argocd
  finalizers:
  - resources-finalizer.argocd.argoproj.io
spec:
  destination:
    namespace: "argocd"
    server: https://kubernetes.default.svc
  project: default
  sources:
```


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```
  - chart: foundation-base-argocd-wrapper
    repoURL: https://gart.software.gevernova.com/artifactory/api/helm/foundation-helm
    targetRevision: BASE-VERSION
    helm:
      valueFiles:
        - $valuesRepo/2024-12/foundation-base-argocd/foundation-base-app-values.yaml
  - repoURL: https://github.build.ge.com/IDMS/hms-deploy.git
    targetRevision: FRA-VERSION
    ref: valuesRepo
  syncPolicy:
    automated:
      prune: false
      selfHeal: true
      allowEmpty: true
    syncOptions:
    - CreateNamespace=false
    - RespectIgnoreDifferences=true
```

## 4.3.10. Foundation Data App Configuration

This section is applicable for ArgoCD based deployment. Specify the address of the Docker Registry and GitHub [GE's Docker Trusted Registry (DTR) or a Nexus registry on a deployment server] in the `foundation-data.yaml`.

In the tar file, the `foundation-data.yaml` file can be found at `apps`.

### 4.3.10.1. foundation-data.yaml

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: foundation-data
  namespace: argocd
  finalizers:
  - resources-finalizer.argocd.argoproj.io
spec:
  destination:
    namespace: "foundation-cluster-zerotrust"
    server: https://kubernetes.default.svc
  project: default
  source:
    helm:
    # valueFiles:
    # - ../../apps-config/global-config/00-foundation-global-values.yaml
    path: 2024-12/foundation-data
    repoURL: https://github.build.ge.com/IDMS/hms-deploy.git
    targetRevision: FRA-VERSION
  syncPolicy:
    automated:
```


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```
      prune: false
      selfHeal: true
    syncOptions:
    - CreateNamespace=false
    - Replace=true
```

## 4.3.11. Foundation OSB App Configuration

This section is applicable for ArgoCD based deployment. Specify the address of the Docker Registry and GitHub [GE's Docker Trusted Registry (DTR) or a Nexus registry on a deployment server] in the `foundation-osb.yaml`.

In the tar file, the `foundation-osb.yaml` file can be found at `apps`.

### 4.3.11.1. foundation-osb.yaml

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: foundation-osb
  namespace: argocd
  finalizers:
  - resources-finalizer.argocd.argoproj.io
spec:
  destination:
    namespace: "foundation-env-default"
    server: https://kubernetes.default.svc
  project: default
  sources:
  - chart: foundation-osb
    repoURL: https://gart.software.gevernova.com/artifactory/api/helm/foundation-helm
    targetRevision: OSB-VERSION
    helm:
      valueFiles:
        - $valuesRepo/2024-12/foundation-helm-profiles/PROFILE/foundation-umbrella-values.yaml
        - $valuesRepo/2024-12/foundation-helm-profiles/PROFILE/global-values.yaml
  - repoURL: https://github.build.ge.com/IDMS/hms-deploy.git
    targetRevision: FRA-VERSION
    ref: valuesRepo
  syncPolicy:
    automated:
      prune: false
      selfHeal: true
      allowEmpty: true
    syncOptions:
    - CreateNamespace=false
    - RespectIgnoreDifferences=true
```


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## 4.3.12. Foundation UI Server App Configuration

This section is applicable for ArgoCD based deployment. Specify the address of the Docker Registry and GitHub [GE's Docker Trusted Registry (DTR) or a Nexus registry on a deployment server] in the `foundation-ui-server.yaml`.

In the tar file, the `foundation-ui-server.yaml` file can be found at `apps`.

### 4.3.12.1. foundation-ui-server.yaml

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: foundation-ui-server
  namespace: argocd
  finalizers:
  - resources-finalizer.argocd.argoproj.io
spec:
  destination:
    namespace: "foundation-env-default"
    server: https://kubernetes.default.svc
  project: default
  sources:
  - chart: foundation-ui-server
    repoURL: https://gart.software.gevernova.com/artifactory/api/helm/foundation-helm
    targetRevision: UI-SERVER-VERSION
    helm:
      releasenName: foundation-ui-server
      valueFiles:
        - $valuesRepo/2024-12/foundation-helm-profiles/PROFILE/foundation-umbrella-values.yaml
        - $valuesRepo/2024-12/foundation-helm-profiles/PROFILE/global-values.yaml
  - repoURL: https://github.build.ge.com/IDMS/hms-deploy.git
    targetRevision: FRA-VERSION
    ref: valuesRepo
  syncPolicy:
    automated:
      prune: false
      selfHeal: true
      allowEmpty: true
    syncOptions:
    - CreateNamespace=false
    - RespectIgnoreDifferences=true
```

## 4.3.13. Hazard Management Kafka App Configuration

This section is applicable for ArgoCD based deployment. Specify the address of the Docker Registry and GitHub [GE's Docker Trusted Registry (DTR) or a Nexus registry on a deployment server] in the `hazard-management-kafka.yaml`.


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In the tar file, the `hazard-management-kafka.yaml` file can be found at `apps`.

### 4.3.13.1. hazard-management-kafka.yaml

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: hazard-management-kafka
  namespace: argocd
  annotations:
    argocd.argoproj.io/sync-wave: "1"
  finalizers:
    - resources-finalizer.argocd.argoproj.io
spec:
  destination:
    namespace: hms
    server: https://kubernetes.default.svc
  project: default
  source:
    helm:
      # parameters:
      # - name: global.registry
      #   value: gart.software.gevernova.com
      valueFiles:
      - ../../apps-config/strimzi-kafka-values.yaml
    path: charts/strimzi-kafka-cluster
    repoURL: https://github.build.ge.com/IDMS/hms-deploy.git
    targetRevision: FRA-VERSION
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
    syncOptions:
      - CreateNamespace=true
      - RespectIgnoreDifferences=true
```

## 4.3.14. Hazard Management Data App Configuration

This section is applicable for ArgoCD based deployment. Specify the address of the Docker Registry and GitHub [GE's Docker Trusted Registry (DTR) or a Nexus registry on a deployment server] in the `hms-data.yaml`.

In the tar file, the `hms-data.yaml` file can be found at `apps`.

### 4.3.14.1. hms-data.yaml

```
apiVersion: argoproj.io/v1alpha1
kind: Application
```


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```
metadata:
  name: hms-data
  namespace: argocd
  finalizers:
  - resources-finalizer.argocd.argoproj.io
spec:
  destination:
    namespace: "hms"
    server: https://kubernetes.default.svc
  project: default
  source:
    helm:
    # valueFiles:
    # - ../../apps-config/global-config/00-foundation-global-values.yaml
    path: 2024-12/hms-data
    repoURL: https://github.build.ge.com/IDMS/hms-deploy.git
    targetRevision: FRA-VERSION
  syncPolicy:
    automated:
      prune: false
      selfHeal: true
    syncOptions:
    - CreateNamespace=false
    - Replace=true
```

## 4.3.15. Hazard Management App Configuration

This section is applicable for ArgoCD based deployment. Specify the address of the Docker Registry and GitHub [GE's Docker Trusted Registry (DTR) or a Nexus registry on a deployment server] in the `hms.yaml`.

In the tar file, the `hms.yaml` file can be found at `apps`.

### 4.3.15.1. hms.yaml

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: hms
  namespace: argocd
  annotations:
    argocd.argoproj.io/sync-wave: "1"
  finalizers:
    - resources-finalizer.argocd.argoproj.io
spec:
  destination:
    namespace: hms
    server: https://kubernetes.default.svc
  project: default
```


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```
  source:
    helm:
      # parameters:
      # - name: global.registry
      #   value: gart.software.gevernova.com
      valueFiles:
      - ../../apps-config/hms-values.yaml
    path: charts/hms
    repoURL: https://github.build.ge.com/IDMS/hms-deploy.git
    targetRevision: FRA-VERSION
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
    syncOptions:
      - CreateNamespace=true
      - RespectIgnoreDifferences=true
```

## 4.3.16. Conflict Manager App Configuration

This section is applicable for ArgoCD based deployment. Specify the address of the Docker Registry and GitHub [GE's Docker Trusted Registry (DTR) or a Nexus registry on a deployment server] in the `conflict-manager.yaml`.

In the tar file, the `conflict-manager.yaml` file can be found at `apps`.

### 4.3.16.1. conflict-manager.yaml

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: conflict-manager
  namespace: argocd
  annotations:
    argocd.argoproj.io/sync-wave: "1"
  finalizers:
    - resources-finalizer.argocd.argoproj.io
spec:
  destination:
    namespace: conflict-manager
    server: https://kubernetes.default.svc
  project: default
  source:
    helm:
      # parameters:
      # - name: global.registry
      #   value: gart.software.gevernova.com
      valueFiles:
      - ../../apps-config/conflict-manager-values.yaml
```


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```
    path: charts/conflict-manager
    repoURL: https://github.build.ge.com/IDMS/conflict-manager-charts.git
    targetRevision: FRA-VERSION
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
    syncOptions:
      - CreateNamespace=true
      - RespectIgnoreDifferences=true
```

## 4.3.17. Preparing AOR Data

Configure the following files:

In this version of Foundation, users can use version 2 or version 3 of the config files related to AOR.

To switch between V2 and V3, use the following config setting in hazard-management-charts/2025-09/foundation-data/values.yaml

aorsManager:

- For v3 aors set below to true

```
    importYamlFileContent: true
    useV3Config: true
    useV3Import: true
```

- For v2 aors set below to true

```
    importFileContent: false
    useV2Config: false
    useAorsXML: false
```

- V2: aors.xml, roles-config.yaml
- V3: aors-config-v3.yaml, roles-config.yaml, aors-import-v3.yaml
- aors.xml: Contains a list of AORs exported from Permit. Used by the data-loader for V2.
- aors-config-v3.yaml: Contains the AORs-to-Group mapping. Used by the data-loader for V3.
- roles-config.yaml: Contains the Roles-to-Group mapping. Used by the data-loader.

For more information about configuring AORs, consult your GE project team.


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## 4.3.18. Hazard Management AOR Configuration Updates

### 4.3.18.1. aors-config-v3.yaml

In the aors-config-v3.yaml file, you'll find configurations specific to AORs for Hazard Management.

- Different types of AORs are supported in v3 like Geographic, DeviceType, Distribution and so on. Each AOR should start with ROOT followed by AOR Type. For example, is ROOT.GEOGRAPHIC....
- AORs are defined under each usergroupName. You can assign access levels such as Read or Write independently for each AOR. In v2, the equivalent values were oms.Read and oms.Write.
- Each AOR can be enabled or disabled using the isDefault flag set to true or false. If isDefault is not provided, the default value is false.

For example:

```
- usergroupName: Supervisors
       selectableAor:
         aors:
           - aorName: ROOT.GEOGRAPHIC.GE.WA
             permissionLevel: ['Read','Write']
             isDefault: true
           - aorName: ROOT.GEOGRAPHIC.GE.OR
             permissionLevel: ['Read','Write']
             isDefault: true
           - aorName: ROOT.GEOGRAPHIC.GE.ORG
             permissionLevel: ['Read', 'Write']
             isDefault: true
           - aorName: ROOT.GEOGRAPHIC.GE.FUNC
             permissionLevel: ['Read', 'Write']
             isDefault: true
           - aorName: ROOT.GEOGRAPHIC.GE.DEFPAREA
             permissionLevel: ['Read', 'Write']
             isDefault: true
- usergroupName: Readonly Viewers
       selectableAor:
         aors:
           - aorName: ROOT.GEOGRAPHIC.GE.WA
             permissionLevel: ['Read']
             isDefault: true
           - aorName: ROOT.GEOGRAPHIC.GE.OR
             permissionLevel: ['Read']
             isDefault: true
           - aorName: ROOT.GEOGRAPHIC.GE.ORG
             permissionLevel: ['Read']
             isDefault: true
```


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### 4.3.18.2. aors-import-v3.yaml

In the aors-import-v3.yaml file, you'll define the AOR hierarchy information for Hazard Management in v3. Here are the key points:

- AOR Hierarchy Configuration: Unlike v2 version, v3 version draws the AOR hierarchy information from this file instead of aors.xml. Ensure all parent and child level AORs are defined in this file.
- ShortName Property: The shortName property in v3 is equivalent to the NameAOR property in aors.xml of v2. If shortName is not provided, the default value would be the last part of an AOR (after the last dot in the name property).

For example:

```
- name: ROOT
   description: Root
   shortName: ROOT
 - name: ROOT.GEOGRAPHIC
   description: GEOGRAPHIC TYPE
   shortName: GEOGRAPHIC
 - name: ROOT.GEOGRAPHIC.GE.WA.GORDON.ROARKE.3LIONSSC.RAVEN
   description: RAVEN
   shortName: RAVEN
 - name: ROOT.GEOGRAPHIC.GE.ORG
   description: ORG
   shortName: ORG
 - name: ROOT.GEOGRAPHIC.GE.ORG.SO_HIGH
   description: SO_HIGH
   shortName: SO_HIGH
 - name: ROOT.GEOGRAPHIC.GE.FUNC
   description: FUNC
   shortName: FUNC
 - name: ROOT.GEOGRAPHIC.GE.FUNC.OUTAGE
   description: OUTAGE
   shortName: OUTAGE
```

### 4.3.18.3. roles-config-v2.yaml

In the roles-config-v2.yaml file, which is used by both v2 version and v3 version, you'll find role configurations.

- There's no change to the existing file unless a role or permission needs to be added or updated.
- When deleting roles in an existing environment and re-deploying, ensure existing AORs are deleted as well because both AORs and roles share the same usergroups.


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## 4.3.19. Preparing a Deployment Folder

> **Note**
>
> Before running deployment procedure below, make sure Foundation and Hazard Management deployment prerequisites are met (Namespaces, root certificates, external credentials secrets, and so on).

On the machine where you deploy the Hazard Management and Conflict Manager solution, prepare a folder for deployment:

1. Create a folder named `apps` (for ArgoCD based deployment) and `apps-config` (for ArgoCD and Helm based deployments) to store the yaml file.
2. Add the Helm chart of Hazard Management by unpacking the chart, hms-deployment-16.18.0.tgz. Update the Docker Registry to either GE's Docker Trusted Registry (DTR) or a Nexus registry if reading the chart from artifactory.
3. Add the Helm chart of Strimzi Kafka by unpacking the chart, strimzi-kafka-cluster-16.18.0.tgz. Update the Docker Registry to either GE's Docker Trusted Registry (DTR) or a Nexus registry if reading the chart from artifactory.
4. Add the Helm chart of Conflict Manager by unpacking the chart, conflict-manager-deployment-16.18.0.tgz. Update the Docker Registry to either GE's Docker Trusted Registry (DTR) or a Nexus registry if reading the chart from artifactory.
5. Copy the Kubernetes cluster's admin.conf file to the "deployment" folder. By default, the admin.conf file is stored in the /etc/kubernetes/ folder.
6. Add the adms Public certificate Authority certificate (ADMS) in YAML file. This certificate is used to sign certificates on the ADMS side. (See Generating Certificates for SSL/TLS Termination with ADMS.)
7. Copy the certificate and key for the Ingress controller and for establishing connections with Active Directory to the "cert" folder. (See Generating Certificates for SSL/TLS Termination with ADMS.)

   > **Note**
   >
   > Ingress certificates are not required for HMS deployment if you already deployed Ingress Controller with enabled TLS termination during the Foundation deployment.

8. Copy the AOR files to the "data-loader" folder.

A sample folder structure that is used for deployment is shown below. This structure can be used as a reference to create the folder structure.

```
<Deployment Folder>
 └──── Foundation helm chart
 └──── Hazard Management helm chart
 └──── Strimzi Kafka helm chart
 └──── Conflict Manager helm chart
```


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```
└──── apps
   └──── custom-namespaces.yaml
   └──── foundation-base.yaml
   └──── foundation-data.yaml
   └──── foundation-osb.yaml
   └──── foundation-ui-server.yaml
   └──── hazard-management-kafka.yaml
   └──── hms-data.yaml
   └──── hms.yaml
   └──── conflict-manager.yaml
└──── apps-config
   └──── hms-values.yaml
   └──── strimzi-kafka-values.yaml
   └──── conflict-manager-values.yaml
```

In the example above, the Foundation Helm chart folder contains all the required files for deploying the Foundation Base, UI, OSB.

## 4.4. Helm Based Deployment

The example command below assumes that digital grid artifactory is used as the Helm repository, with the corresponding user and password stored in the environment variables ${GRID_ARTIFACTORY_CREDS_USR} and ${GRID_ARTIFACTORY_CREDS_PSW}.

In air-gapped environments, replace https://gart.software.gevernova.com/artifactory/api/helm/adms-helm with the URL of your local Helm repository, and provide the local repository's username and password accordingly.

```
helm repo add adms-helm https://gart.software.gevernova.com/artifactory/api/helm/adms-helm --username
${GRID_ARTIFACTORY_CREDS_USR} --password ${GRID_ARTIFACTORY_CREDS_PSW}

curl -sSL https://storage.googleapis.com/kubernetes-
release/release/v${KUBECTL_VERSION}/bin/linux/amd64/kubectl -o kubectl && chmod +x kubectl || true

# create image pull secret
./kubectl create secret docker-registry ${GRIDART_SECRET_NAME} --docker-server=${DGA_REGISTRY}
--docker-username=${GRID_ARTIFACTORY_CREDS_USR} --docker-password=${GRID_ARTIFACTORY_CREDS_PSW} -n
"${HMS_DEPLOYMENT_NAMESPACE}" || true

./kubectl patch serviceaccount default -n "${HMS_DEPLOYMENT_NAMESPACE}"  -p '{\"imagePullSecrets\":
[{\"name\": \"docker-registry\"}]}'
  # create PG secret

./kubectl get secret postgres-db-pg-cluster-postgres-secret -n "${PLATFORM_NAMESPACE}" -o yaml | grep
-v '^\\s*namespace:\\s' | kubectl apply -n "${HMS_DEPLOYMENT_NAMESPACE}" -f - || true

#Export Variables
export VALUES_ARGS="--values hms-values.yaml
            --set global.zitadel.enabled=true
            --set global.clusterExternalUrl=https://<Cluster_External_DNS>"
```


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```
helm upgrade -i <release-name> adms-helm/<umbrella-chart-name> --version <umbrella-chart-version>
--wait $VALUES_ARGS
```

## 4.5. ArgoCD Based Deployment

> **Note**
>
> Unlike the helm-only use case, ArgoCD deployment requires pushing the configuration manifests to a Git repository additionally. In either case, make a note of the location of each configuration manifest in Git, as they will be used to configure the HMS ArgoCD application.

The command below assumes that your kubectl client is configured with the argocd cluster kubeconfig file, meaning you have access to it. If not, please contact the argocd admin user/team to obtain the argocd credentials and log in using the classical way.

```
kubectl config set-context --current --namespace=argocd

argocd login --core
```

- add helm repository

  The example command below assumes digital grid artifactory is used as the helm repository, and that the corresponding user and password are stored in respective environment variables ${GRID_ARTIFACTORY_CREDS_USR} and ${GRID_ARTIFACTORY_CREDS_PSW}

  In airgap environments, replace https://gart.software.gevernova.com/artifactory/api/helm/adms-helm with the url to your local helm repository. And pass in the local helm repo's username and password accordingly.

  ```
  helm repo add adms-helm https://gart.software.gevernova.com/artifactory/api/helm/adms-helm
  --username ${GRID_ARTIFACTORY_CREDS_USR} --password ${GRID_ARTIFACTORY_CREDS_PSW}

  curl -sSL https://storage.googleapis.com/kubernetes-
  release/release/v${KUBECTL_VERSION}/bin/linux/amd64/kubectl -o kubectl && chmod +x kubectl || true

  # Foundation Base
  argocd app create -f foundation-base.yaml \
    --helm-set-string
  spec.sources.values.adhocGlobals.global.clusterExternalUrl=https://${CLUSTER_EXTERNAL_DNS} \
    --helm-set-string spec.sources.values.foundationProfile.name=${foundationProfile} \
    --upsert
  argocd app wait foundation-base --sync
  argocd app wait foundation-base --operation
  argocd app wait foundation-base --health

  # Foundation Data
  argocd app create -f foundation-data.yaml \
    --helm-set-string
  spec.sources.values.adhocGlobals.global.clusterExternalUrl=https://${CLUSTER_EXTERNAL_DNS} \
  ```


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```
  --helm-set-string spec.sources.values.foundationProfile.name=${foundationProfile} \
  --upsert
argocd app wait foundation-data --sync
argocd app wait foundation-data --operation
argocd app wait foundation-data --health

# Foundation OSB
argocd app create -f foundation-osb.yaml \
  --helm-set-string global.clusterExternalUrl=https://${CLUSTER_EXTERNAL_DNS} \
  --upsert
argocd app wait foundation-osb --sync
argocd app wait foundation-osb --operation
argocd app wait foundation-osb --health

# Hazard Management Kafka
argocd app create -f  hazard-management-kafka.yaml --repo https://github.build.ge.com/IDMS/hms-
deploy.git \
  --helm-set-string global.clusterExternalUrl=https://${CLUSTER_EXTERNAL_DNS} \
  --upsert
argocd app wait hazard-management-kafka --sync
argocd app wait hazard-management-kafka --operation
argocd app wait hazard-management-kafka --health

# Foundation UI Server
argocd app create -f foundation-ui-server.yaml \
  --helm-set-string global.clusterExternalUrl=https://${CLUSTER_EXTERNAL_DNS} \
  --upsert
argocd app wait foundation-ui-server --sync
argocd app wait foundation-ui-server --operation
argocd app wait foundation-ui-server --health

# create image pull secret
kubectl create secret docker-registry ${GRIDART_SECRET_NAME} --docker-server=${DGA_REGISTRY}
--docker-username=${GRID_ARTIFACTORY_CREDS_USR} --docker-password=${GRID_ARTIFACTORY_CREDS_PSW} -n
"${HMS_DEPLOYMENT_NAMESPACE}" || true

kubectl patch serviceaccount default -n "${HMS_DEPLOYMENT_NAMESPACE}"  -p '{\"imagePullSecrets\":
[{\"name\": \"docker-registry\"}]}'
# create PG secret

kubectl get secret postgres-db-pg-cluster-postgres-secret -n "${PLATFORM_NAMESPACE}" -o yaml |
grep -v '^\\s*namespace:\\s' | kubectl apply -n "${HMS_DEPLOYMENT_NAMESPACE}" -f - || true

argocd repo add https://gart.software.gevernova.com/artifactory/api/helm/adms-helm \
  --type helm \
  --name adms-helm \
  --username ${GRIDART_USR} \
  --password ${GRIDART_PSW} \
  --upsert

# HMS Data
argocd app create -f hms-data.yaml \
 --helm-set-string
```


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```
spec.sources.values.adhocGlobals.global.clusterExternalUrl=https://${CLUSTER_EXTERNAL_DNS} \
 --helm-set-string spec.sources.values.foundationProfile.name=${foundationProfile} \
 --upsert
argocd app wait hms-data --sync
argocd app wait hms-data --operation
argocd app wait hms-data --health

# HMS
argocd app create -f hms.yaml --repo https://github.build.ge.com/IDMS/hms-deploy.git \
--helm-set-string global.clusterExternalUrl=https://${CLUSTER_EXTERNAL_DNS} \
--helm-set-string global.zitadel.enabled=true \
--upsert

argocd app wait hms --sync --timeout 600
argocd app wait hms --operation --timeout 600
argocd app wait hms --health --timeout 600

# create image pull secret
kubectl create secret docker-registry ${GRIDART_SECRET_NAME} --docker-server=${DGA_REGISTRY}
--docker-username=${GRID_ARTIFACTORY_CREDS_USR} --docker-password=${GRID_ARTIFACTORY_CREDS_PSW} -n
"${CONFLICT_MANAGER_DEPLOYMENT_NAMESPACE}" || true

kubectl patch serviceaccount default -n "${CONFLICT_MANAGER_DEPLOYMENT_NAMESPACE}"  -p
'{\"imagePullSecrets\": [{\"name\": \"docker-registry\"}]}'

# Conflict Manager
argocd app create -f conflict-manager.yaml --repo https://github.build.ge.com/IDMS/conflict-
manager-charts.git \
  --helm-set-string global.clusterExternalUrl=https://${CLUSTER_EXTERNAL_DNS} \
  --helm-set-string global.zitadel.enabled=true \
  --upsert
argocd app wait conflict-manager --sync
argocd app wait conflict-manager --operation
argocd app wait conflict-manager --health
```

**Important**

After deploying Foundation, HMS, Conflict Manager Apps, restart the ADMS interface servers (DMS API Server, OMS Interface Server, DMS Interface Server and NetViewer Adapter).


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# 5. Configuring ADMS

## 5.1. Configuring the ingress.local Path

> **Note**
>
> Skip this step if you have already performed the configuration on the DNS server and your hosts already have DNS names.

**To enable access to your Kubernetes master IP address using the ingress.local alias:**

1. Open the hosts file (`C:\Windows\System32\drivers\etc\hosts`) using a text editor that is run with Administrator rights.
2. Add the following entry to the hosts file:

   ```
   <Kubernetes master IP address> ingress.local service.ingress.local zitadel.ingress.local
   admin.ingress.local a6dashboard.ingress.local
   ```

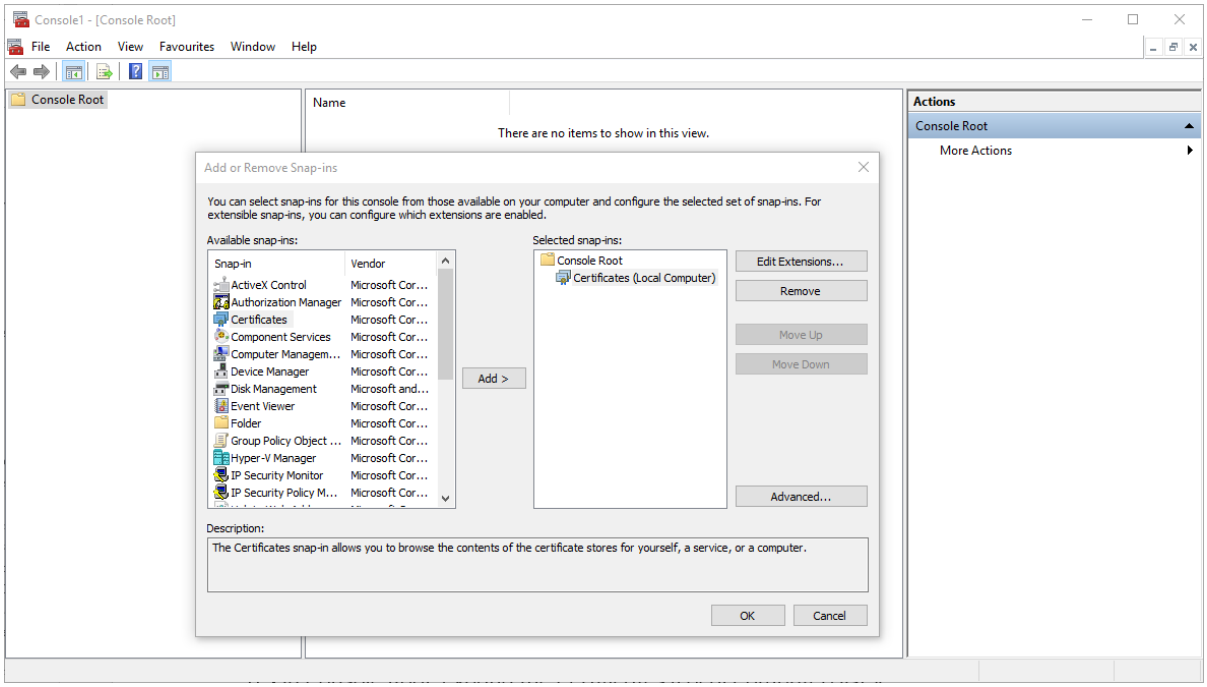
   After this configuration, you can use ingress.local instead of your Kubernetes host name in all configurations. After migrating to Zitadel UAA, the URL changes to https://zitadel.ingress.local/oauth/v2/token and the service is accessed via https://service.ingress.local.

## 5.2. Importing Certificates to the ADMS Hosts

> **Important**
>
> Perform the following steps on each host where DMS API Server, OMS Interface Server, NetViewer Adapter, Tag REST service and Hazard Management DS services are running.

**To import certificates to an ADMS interface server host:**

1. Copy the Root certificate authority certificate (`rootCA.crt`) and the ADMS server PFX file with a certificate and the associated private key (`adms-server.pfx`) to the ADMS machine.
2. Open the Microsoft Management console.
3. From the File menu, select Add/Remove Snap-in.
4. From the available snap-ins, select Certificates and click Add.
5. Select the Computer Account option and click Next.
6. Select the Local Computer option and click Finish.




7. Click OK.
8. In Console Root, expand the Certificates (Local Computer) list.
9. Right-click Personal, select All Tasks, and click Import. The Certificate Import wizard appears.
10. Verify that the Store location is Local Machine and click Next.
11. Click Browse, select the rootCA.crt file to import, and click Next.
12. Select the Place All Certificates in the Following Store option, select Personal, and click Next.
13. Click Finish.

    The root certificate appears in the Certificates list.

14. Perform the same operations to import the ADMS PFX file with a certificate and the associated private key (`adms-server.pfx`). Enter the password that you used when generating the certificate (see Generating Certificates for SSL/TLS Termination with ADMS).

    > **Note**
    >
    > To browse for a .pfx file, select the Personal Information Exchange or All Files filter in the Open dialog box.

15. Verify that the Root certificate authority certificate (HMS Root CA) and the ADMS server certificate (ADMS) are listed in the Personal folder under Certificates (Local Computer).
16. Right-click the ADMS server certificate (ADMS), point to All Tasks, and click Manage Private Keys.

    Alternatively, open the Security tab of the certificate Properties display.

17. Add the user or service account that is used to run NetViewer Adapter (or select Everyone,

which is less secure) and give Read permissions.

> **Important**
>
> Make sure you are giving permissions to the private key, not the public key.

This allows access to the private key of the imported certificate to the user (service account) that is used to run the NetViewer Adapter. Otherwise, the NetViewer Adapter is not able to serve data in HTTPS mode.

18. Restart the NetViewer Adapter if it was already running.
19. In the Microsoft Management Console, copy the Root certificate authority certificate (HMS Root CA) and the ADMS server certificate (ADMS) certificate using the context menu.
20. Navigate to the Certificates > Trusted Root Certification Authorities > Certificates folder.
21. Paste the HMS Root CA and ADMS certificates using the context menu.
22. Add or copy the HMS Root CA and ADMS certificates to the Certificates - Current User store.

    This step is optional, but should be done if you do not bind a certificate to a port as described further.

## 5.3. Binding Certificates to the ADMS Interface Servers

> **Important**
>
> Perform the following steps on each server where DMS API Server, OMS Interface Server, Tag REST Service and Hazard Management DS services are running.

**To bind certificates to the ADMS interface servers**:

1. In the Microsoft Management Console, double-click the ADMS certificate.

   The certificate properties window appears.

2. On the Details tab, copy the Thumbprint value.

   > **Important**
   >
   > Delete all spaces in the copied string, if any.

3. Run the following command to bind the ADMS SSL certificate to the DMS API Server (50058), OMS Interface Server(8812), Tag REST Service(38669), Hazard Management DS(30002):

```
netsh http>add sslcert ipport=0.0.0.0:8812 certhash=<Cert_Thumbprint> appid=<Application_GUID>
netsh http>add sslcert ipport=0.0.0.0:50058 certhash=<Cert_Thumbprint> appid=<Application_GUID>
netsh http>add sslcert ipport=0.0.0.0:38669 certhash=<Cert_Thumbprint> appid=<Application_GUID>
netsh http>add sslcert ipport=0.0.0.0:30002 certhash=<Cert_Thumbprint> appid=<Application_GUID>
```

> **Note**
>
> This action fails if a certificate is already bound to the specified port.

4. Verify the certificate port binding configuration using the following command:

```
netsh http> show sslcert
```

# 5.4. Configuring Main Servers

> **Important**
>
> Perform the following steps on each server where the referenced applications are running.

> **Note**
>
> It is assumed that the ADMS Server and ADMS Interfaces kits are already deployed, and the HMS and Foundation ports are open in Windows firewall.

## Configuring the NetServerDnafConfig.txt file

- DMS API Server connects to DNAF server through gRPC. Ensure that the below lines are uncommented.

```
GRPC_SERVICE_PORT   = 50055
DMS_ENABLE_GRPC_API_SERVICE = 1
```

## Configuring the NetViewerAdapterConfig.xml file

1. Enable a secure web service:

```
<SecureWS>true</SecureWS>
```

2. Set certificate subject matching O=GE of the ADMS CRT certificate (adms-server.crt/adms-server.pfx):

```
<CertificateSubject>GE</CertificateSubject>
```

3. Specify the listening port.

```
<Port>24468</Port>
```

4. Disable Common Notification Framework:

```
<DisableCnf>true</DisableCnf>
```

5. Enable pushing tiles to the Tile service.

```
<EnablePublishingToTileService>true</EnablePublishingToTileService>
```

6. Set the Tile service compression to gzip.

```
<TileServiceCompression>gzip</TileServiceCompression>
```

7. Set a secured connection to the Tile service:

```
<TileServiceUrl>wss://service.ingress.local/tilecacheservice</TileServiceUrl>
```

8. Set a zitadel UAA endpoint:

```
<UaaTokenServer>https://zitadel.ingress.local/oauth/v2/token</UaaTokenServer>
```

9. Create the following system environment variables to provide the Zitadel access token required by the NetViewer Adapter. You can obtain these values from the Kubernetes deployment:

```
   NETVIEW_UAA_AUTH_PROJECTID
   NETVIEW_UAA_AUTH_CLIENTID
   NETVIEW_UAA_AUTH_CLIENTSECRET

# Get the Zitadel Project ID
kubectl get configmap -n foundation-cluster-zerotrust zitadel-projects -o
jsonpath='{.data.foundation\.foundation\.id}'

# Get NetView UAA Client ID and Secret
kubectl get secret netview-adapter-client-secret -n hms -o jsonpath='{.data.clientId}' | base64
--decode
kubectl get secret netview-adapter-client-secret -n hms -o jsonpath='{.data.clientSecret}' |
base64 --decode
```

## Configuring the OmsInterfaceServerConfig.xml file

1. Check the default port values:

```
<WebServiceBaseURL>http://<Main_Server_Host>:8812/</WebServiceBaseURL>
```

2. Enable the secure web service:

```
<SecureWebService>true</SecureWebService>
```

## Configuring the ApiServerConfig.xml file

1. Enable the secure web service:

```
<SecureWebService>true</SecureWebService>
```

## Configuring the appsettings.json file of DMS API Service

1. Update the kestrel configuration:

```
{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*",
  "Kestrel": {
        "Endpoints": {
            "Http": {
                "Url": "https://*:50058"
            }
        }
    }
}
```

# 6. Validating the HMS Setup

**To validate the correct operation of the HMS system:**

1. Run HMS pods and services in Kubernetes.
2. Start the core ADMS servers (DMS server, OMS server, DNAF Server, Call server, Crew server, and so on).
3. Start the ADMS NetViewer Adapter (or restart the application if it is already running to apply changes).
4. Start the ADMS OMS Interface Server.
5. Start the DMS API Server.
6. Start the Tag REST API Service
7. Navigate to https://service.<k8s-cluster-url>/hms.
8. Log in to HMS.
9. Verify that HMS displays the correct geographic network view, layers, components, and so on under NetworkViewer tab.
10. Create a Hazard event and ensure you can search for stations and Feeders.
11. Create another Hazard event and ensure that both are displayed in the event list.

# 7. Maintenance and Troubleshooting

## 7.1. Using the Developer Console

Use the developer console in a web browser for application debugging.

To open the developer console in Google Chrome, press Ctrl+Shift+J.

## 7.2. Network data

**Problem:** Network Viewer does not show the network data

**Solution:** Verify that ADMS servers that communicate with HMS are correctly configured, running, and connected to the HMS Kubernetes cluster.

- Verify that OMS Interface Server, DMS API Server, and NetViewer Adapter are running and check their log files for errors.
- Verify that the "TileServiceClient websocket connection has been OPENED." message appears in the NetViewer Adapter log.

## 7.3. Search for Station and Feeder data

**Problem:** While creating the event, entering station or feeder details does not show any results.

**Solution:** Verify that the DMS API Server is correctly configured, running, and accessible from the kubernetes cluster.

## 7.4. Events creation

**Problem:** Unable to create an event.

**Solution:** Review logs from the Hazard Management pod and Hazard Management DS service for exceptions. Ensure that Postgres is running and accessible to the Hazard Management Ds service.

## 7.5. Error: HTTP Could Not Register URL

**Problem:** The following OMS Interface Server exception appears:

```
HTTP could not register URL http://+:8812/oms/services/. Another application has already registered
this URL+++
```

**Explanation:** In the default configuration, the OMS Interface Server listens at http://+:8812/oms.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Listening at a particular HTTP address requires administrator privileges or account permissions to reserve the URL.

The error could occur due to one of the following reasons:

- There are no proper rights.
- The port is registered by another application.
- The port is registered with https (not http).

**Solution 1:**

Run the OMS Interface server as an administrator.

**Solution 2:**

Use `Netsh.exe` to give the account permissions to reserve the URL.

**To check and update the system HTTP URL namespace reservations:**

1. Run Windows PowerShell as Administrator.
2. Run the "netsh" command.

   `netsh`

3. Run the "http" command. `http`
4. Run one of the following commands to check the current URL assignments in the system:

   - `show urlacl`
   - `show urlacl | findstr /I 880`

   The list may show whether another URL has http or https binding at the same port number (8812).

5. Run the following command to remove an unwanted URL assignment:

   `delete urlacl URL=<URL>`

   For example:

   `http delete urlacl URL=http://+:8812/`

   Or:

   `http delete urlacl URL=https://+:8812/`

6. Run the following command to re-add a new URL assignment for the OMS Interface Server:

   ```
   add urlacl URL=http://+:8812/ user="NT Network Service"
   ```

# 7.6. Chrome Exits when the System is under Load

**Problem:** When the system is under high load, a message saying "Aw Snap, something went

wrong" appears and Chrome exits.

**Reason:** This issue can occur when using the 32-bit version of Chrome.

**Solution:** Use the 64-bit version of Chrome.

# 8. Server Configuration Changes

## 8.1. Customer Experience Adapter

> **Note**
>
> In the NewValue tag, the user must set the PlannedOutageType value to "3" in order to receive updates for planned outages related to hazard events.

```
<Action>
<Name>CreatePlannedOutage</Name>
<Enabled>true</Enabled>
<Filter>
<FieldName>PlannedOutageType</FieldName>
<NewValue>3</NewValue>
</Filter>
</Action>
```

```
<Action>
<Name>UpdatePlannedOutage</Name>
<Enabled>true</Enabled>
<Filter>
<FieldName>PlannedOutageType</FieldName>
<NewValue>3</NewValue>
</Filter>
</Action>
```

```
<Action>
<Name>DeletePlannedOutage</Name>
<Enabled>true</Enabled>
<Filter>
<FieldName>PlannedOutageType</FieldName>
<NewValue>3</NewValue>
</Filter>
</Action>
```


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